

BFS Traversal

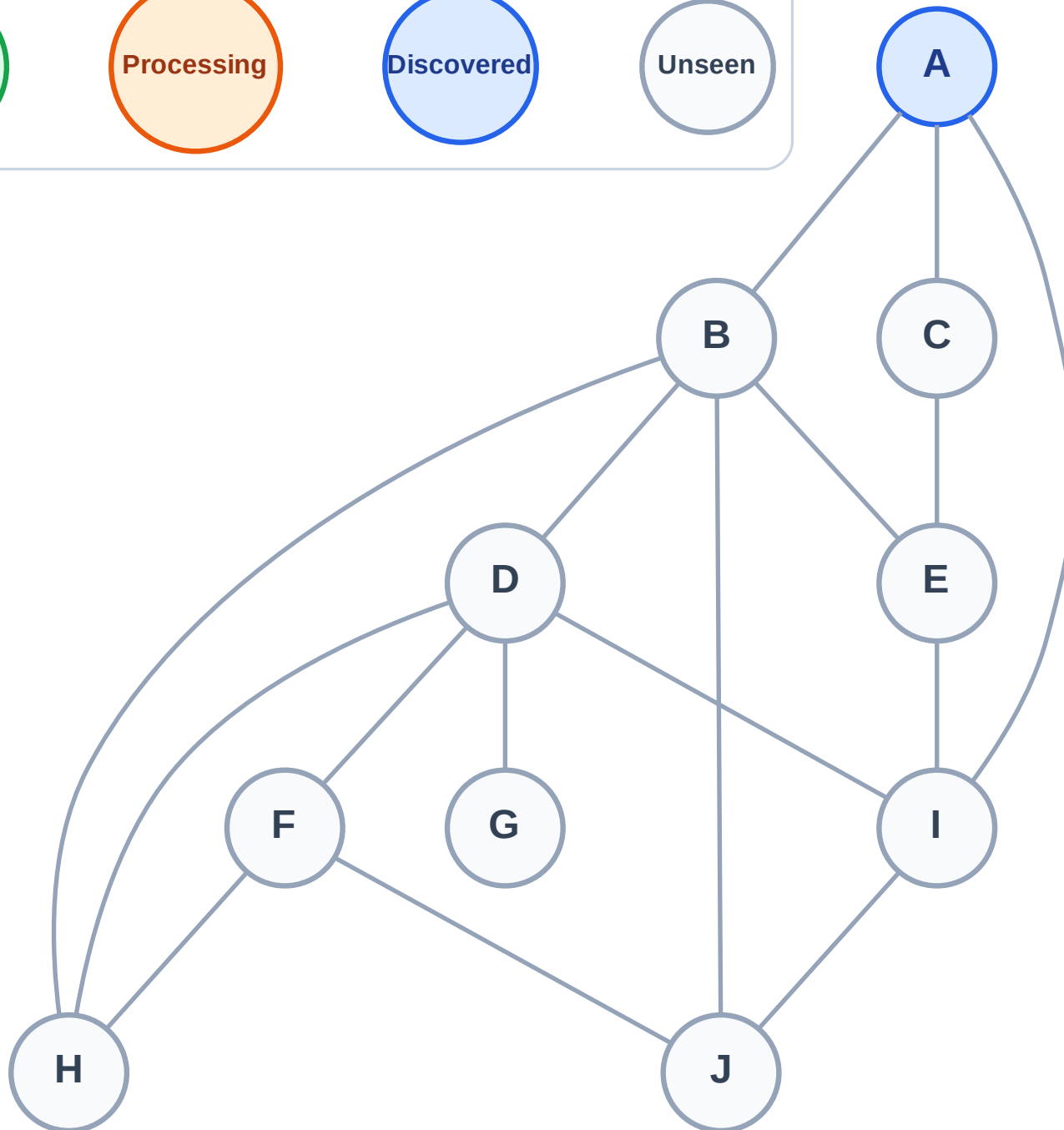
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

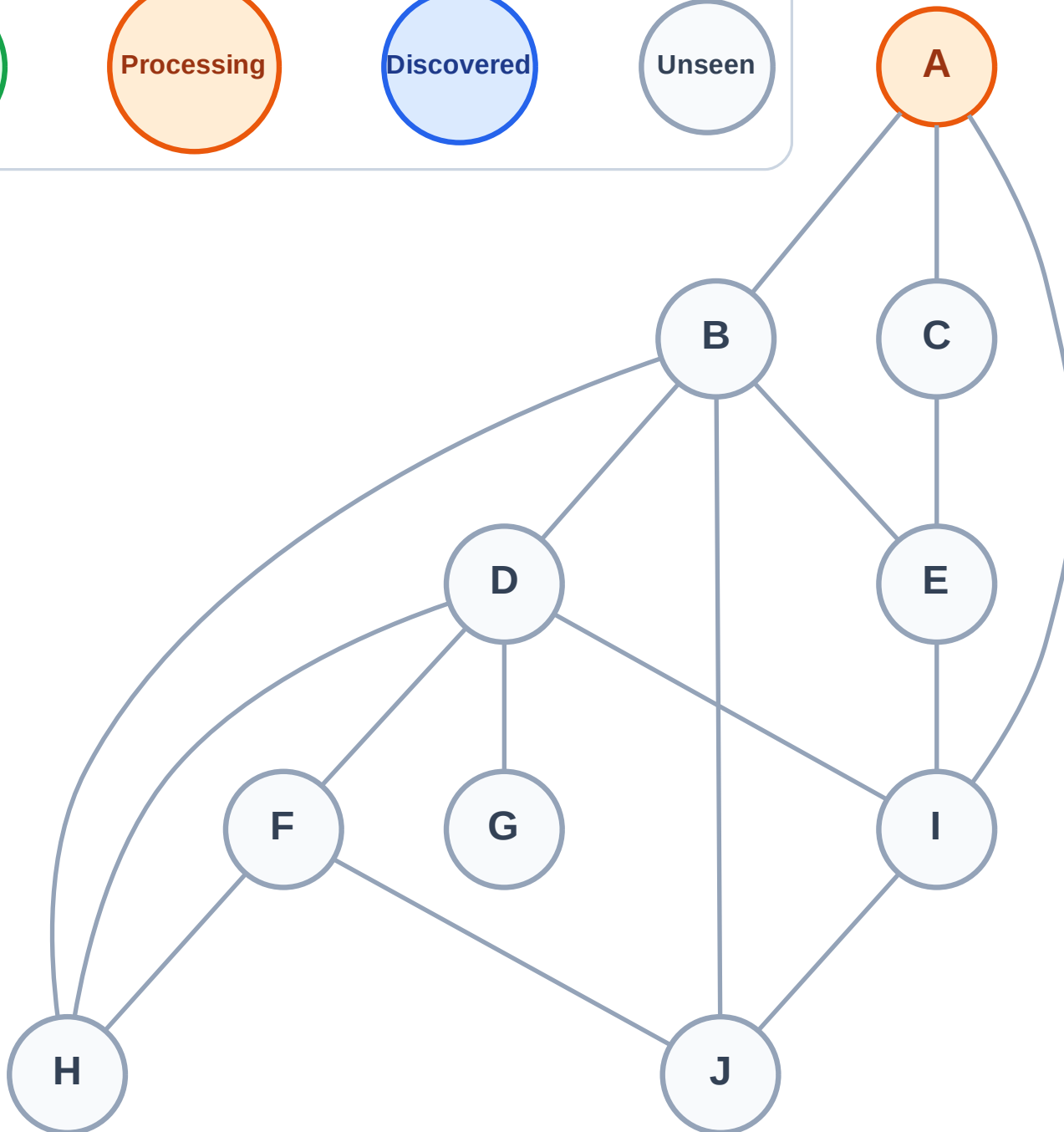
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

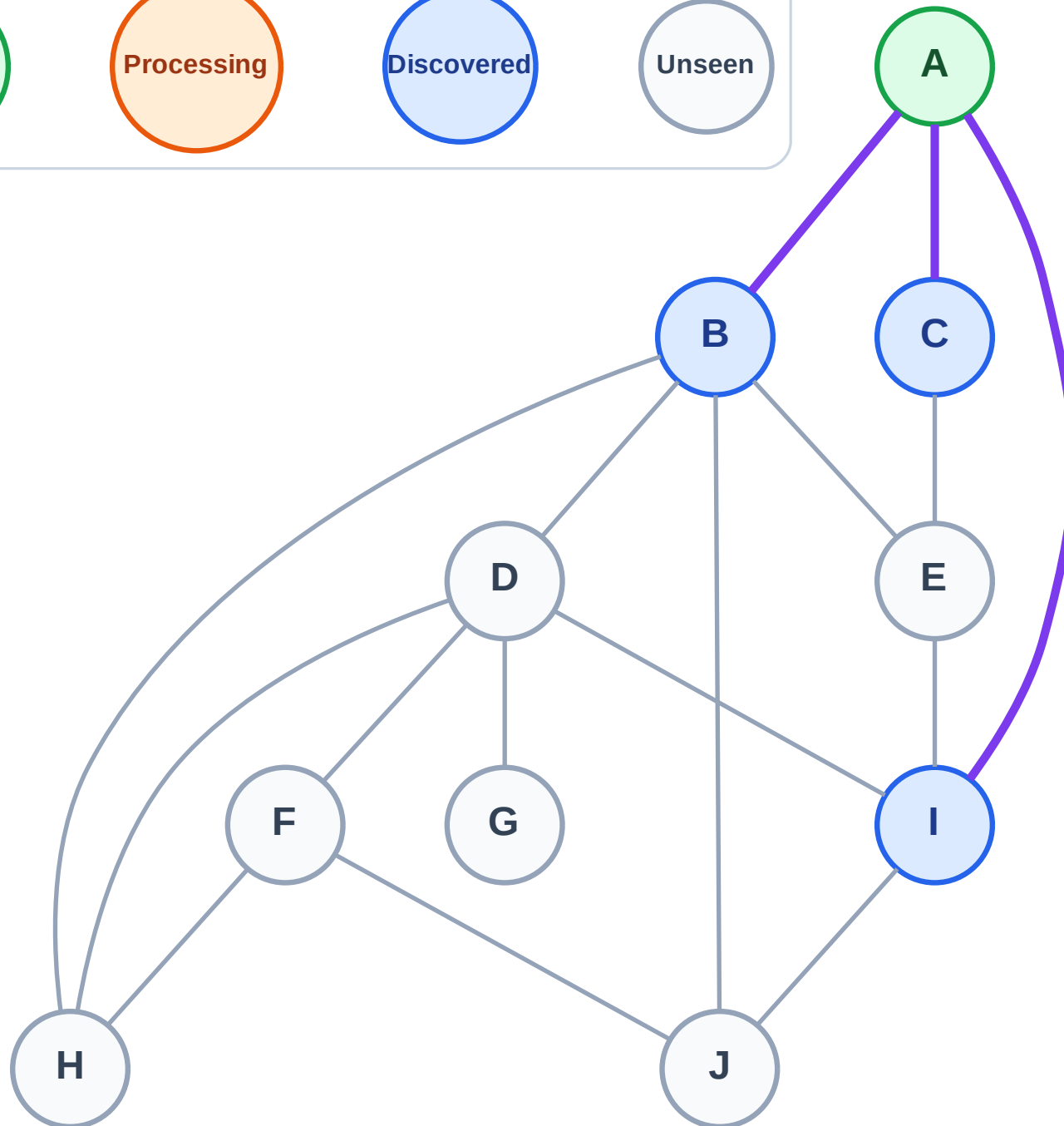
Step 3: Discover B, C, I from A

Legend



Queue: B → C → I

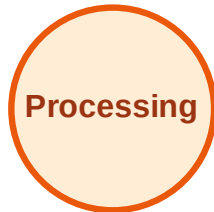
Visited: A



BFS Traversal

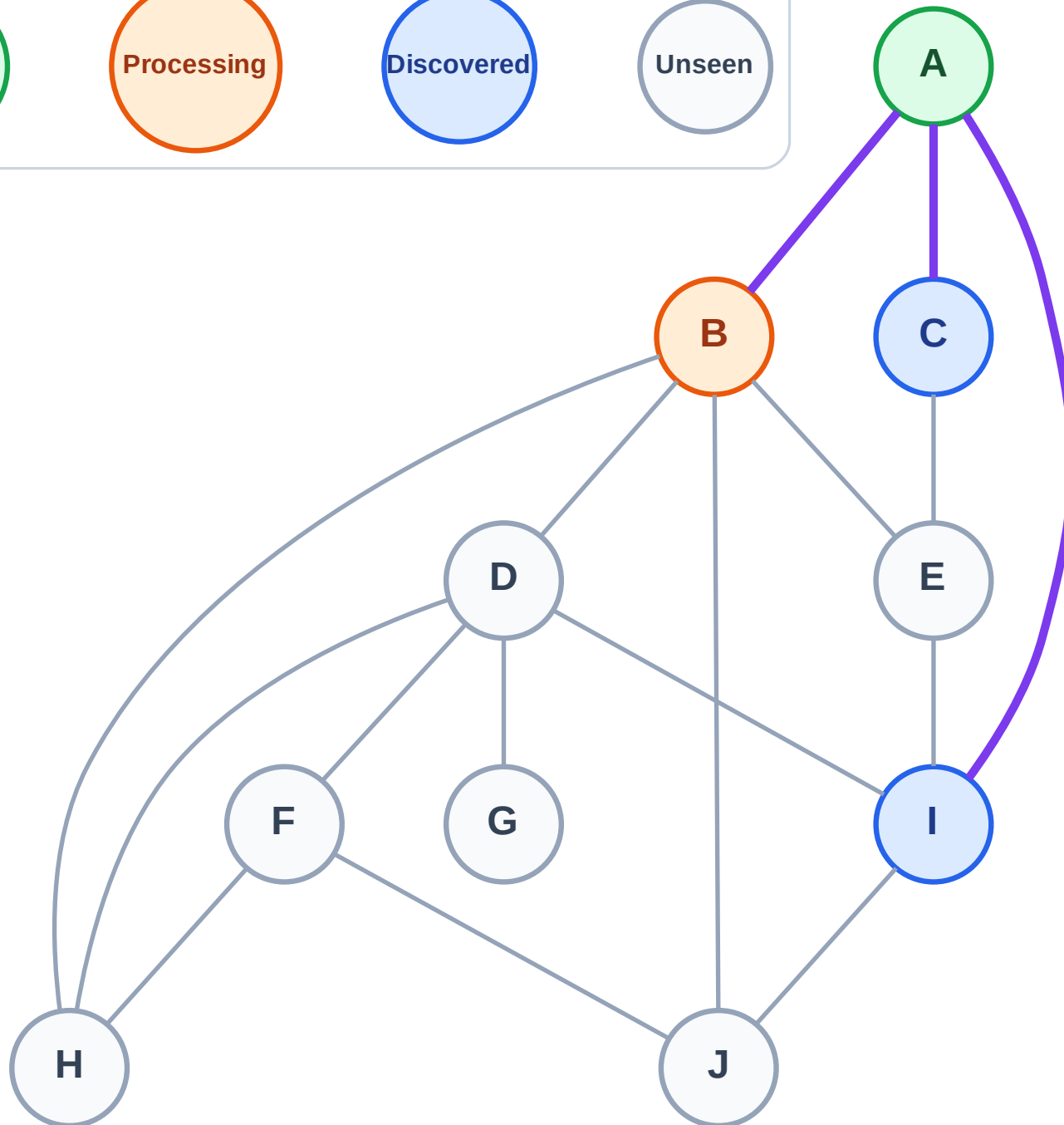
Step 4: Process node B

Legend



Queue: C → I

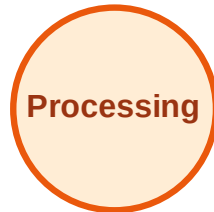
Visited: A



BFS Traversal

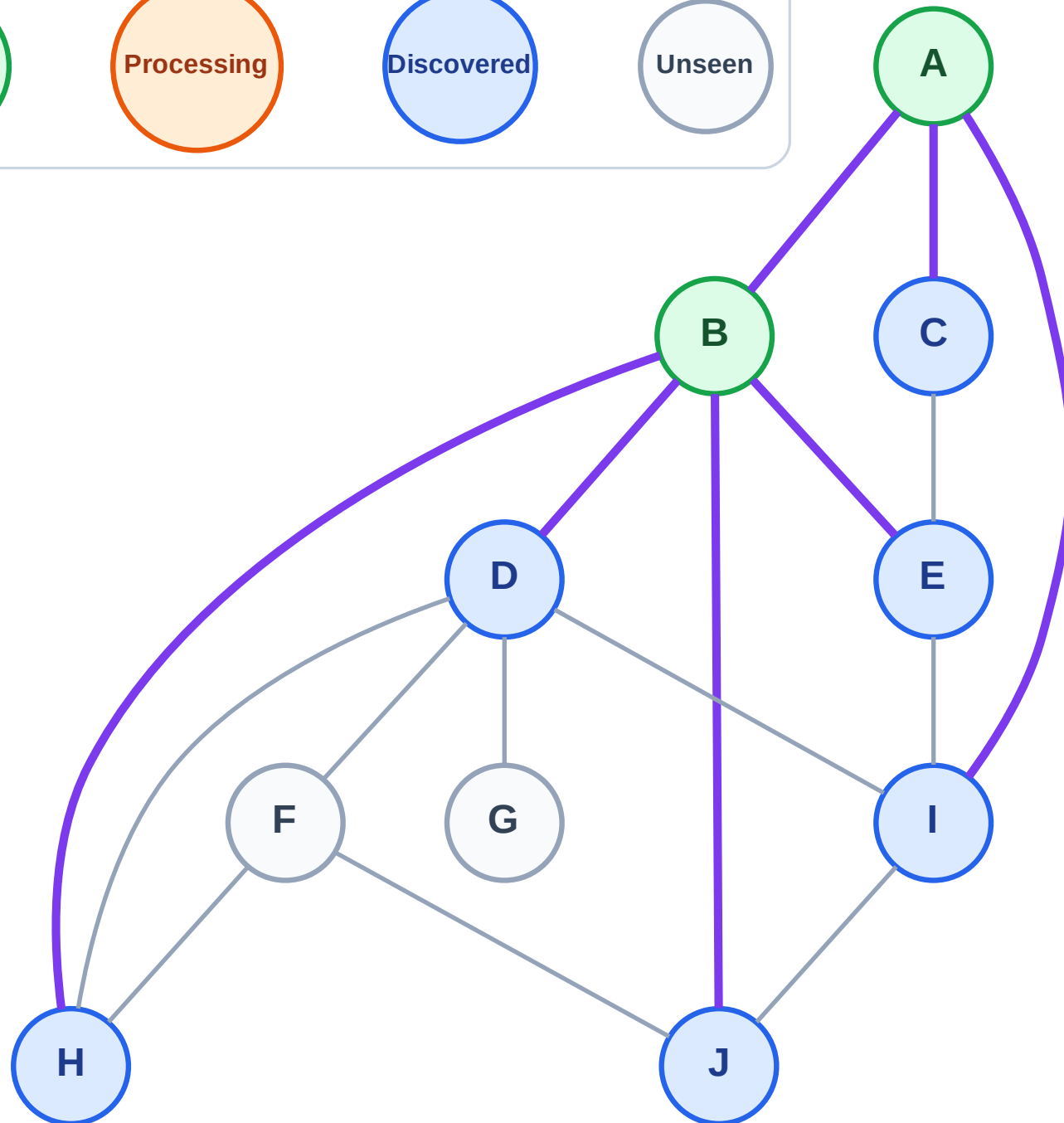
Step 5: Discover D, E, H, J from B

Legend



Queue: C → I → D → E → H → J

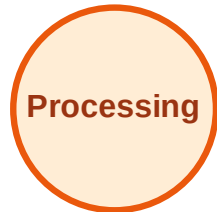
Visited: A, B



BFS Traversal

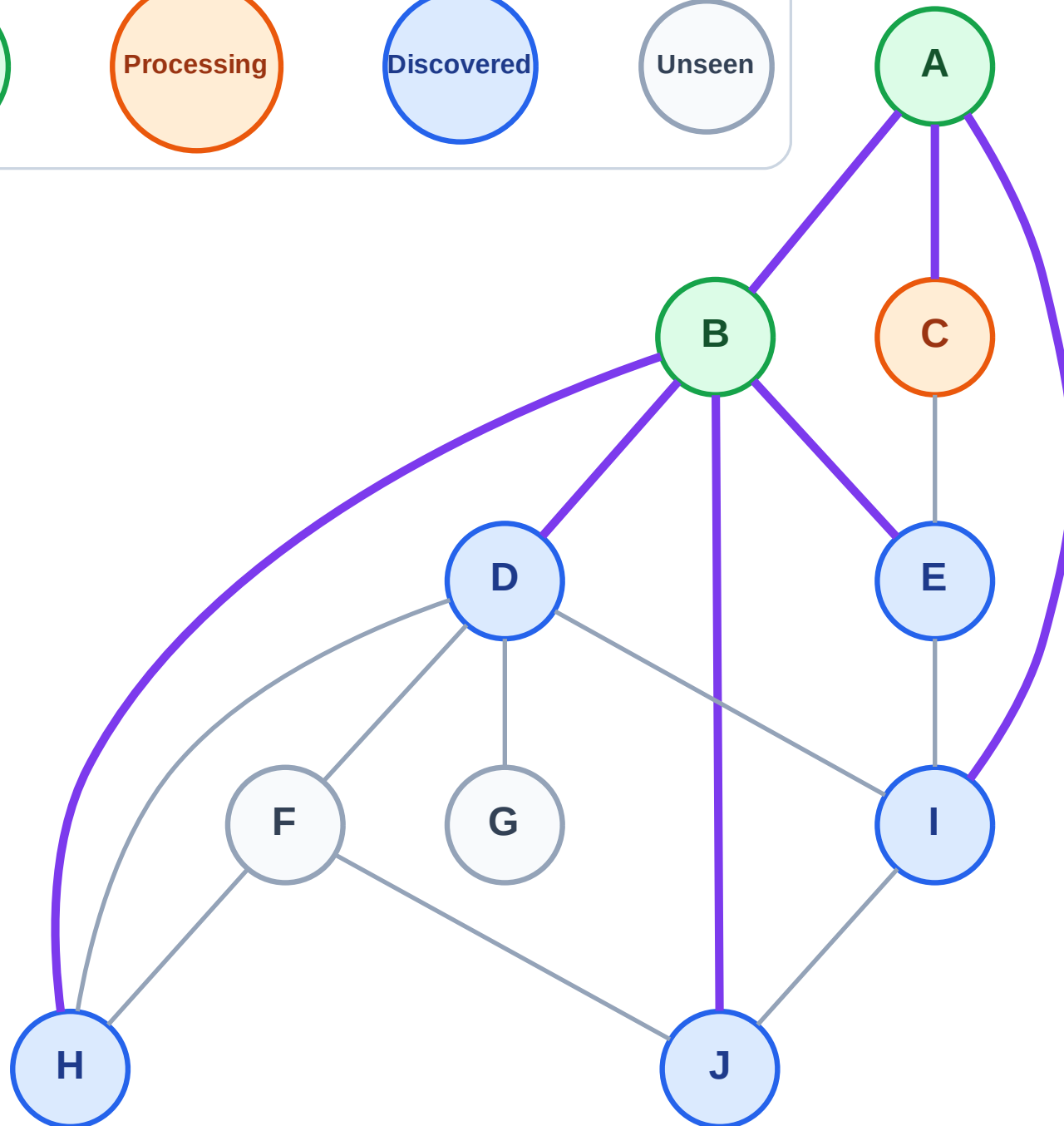
Step 6: Process node C

Legend



Queue: I → D → E → H → J

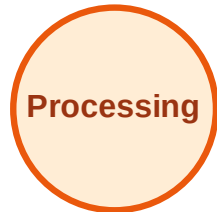
Visited: A, B



BFS Traversal

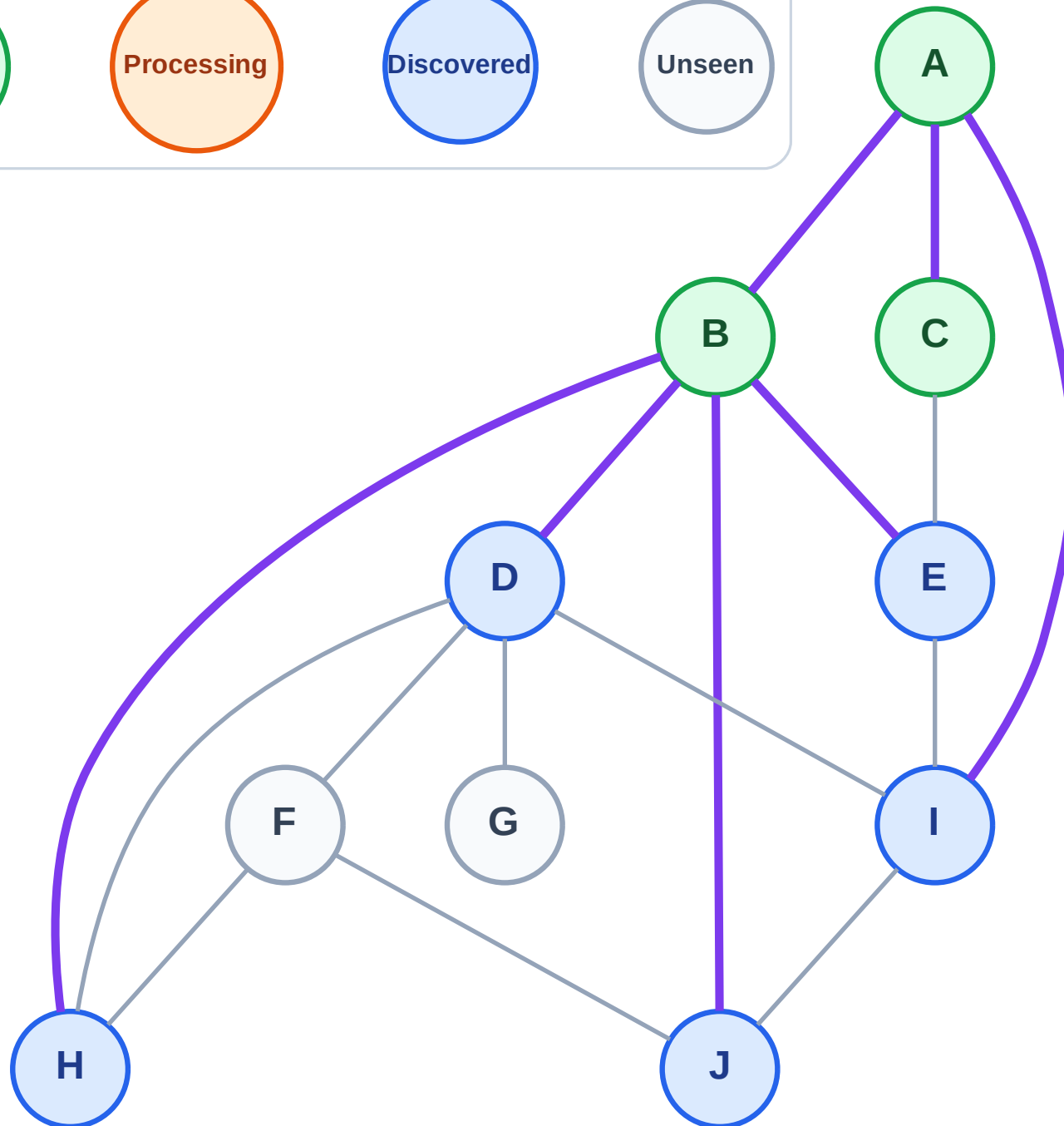
Step 7: No new neighbors from C

Legend



Queue: I → D → E → H → J

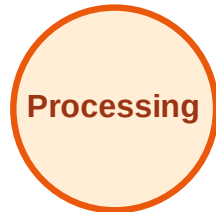
Visited: A, B, C



BFS Traversal

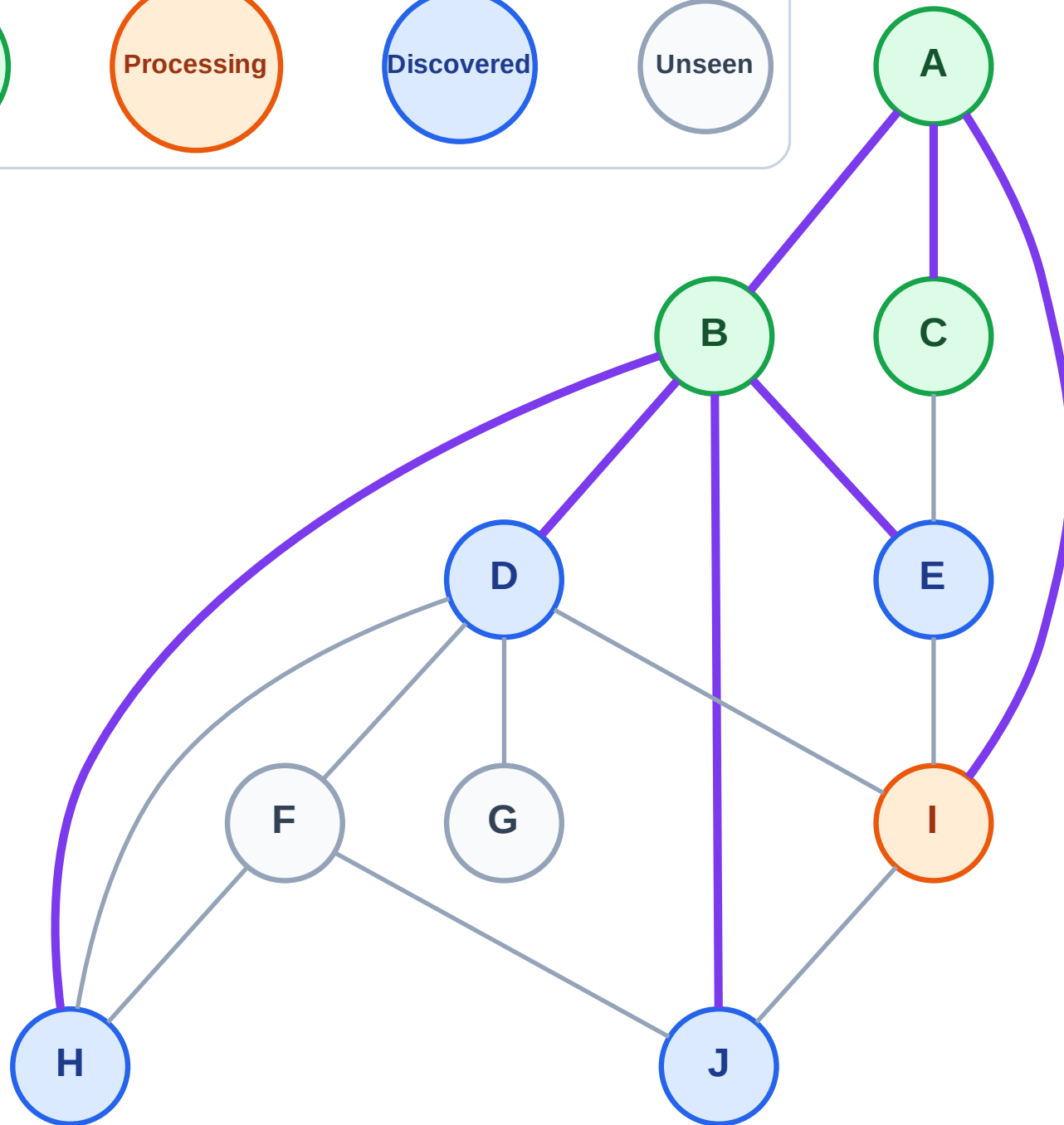
Step 8: Process node I

Legend



Queue: D → E → H → J

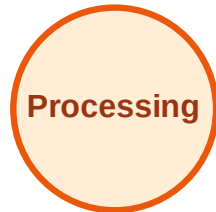
Visited: A, B, C



BFS Traversal

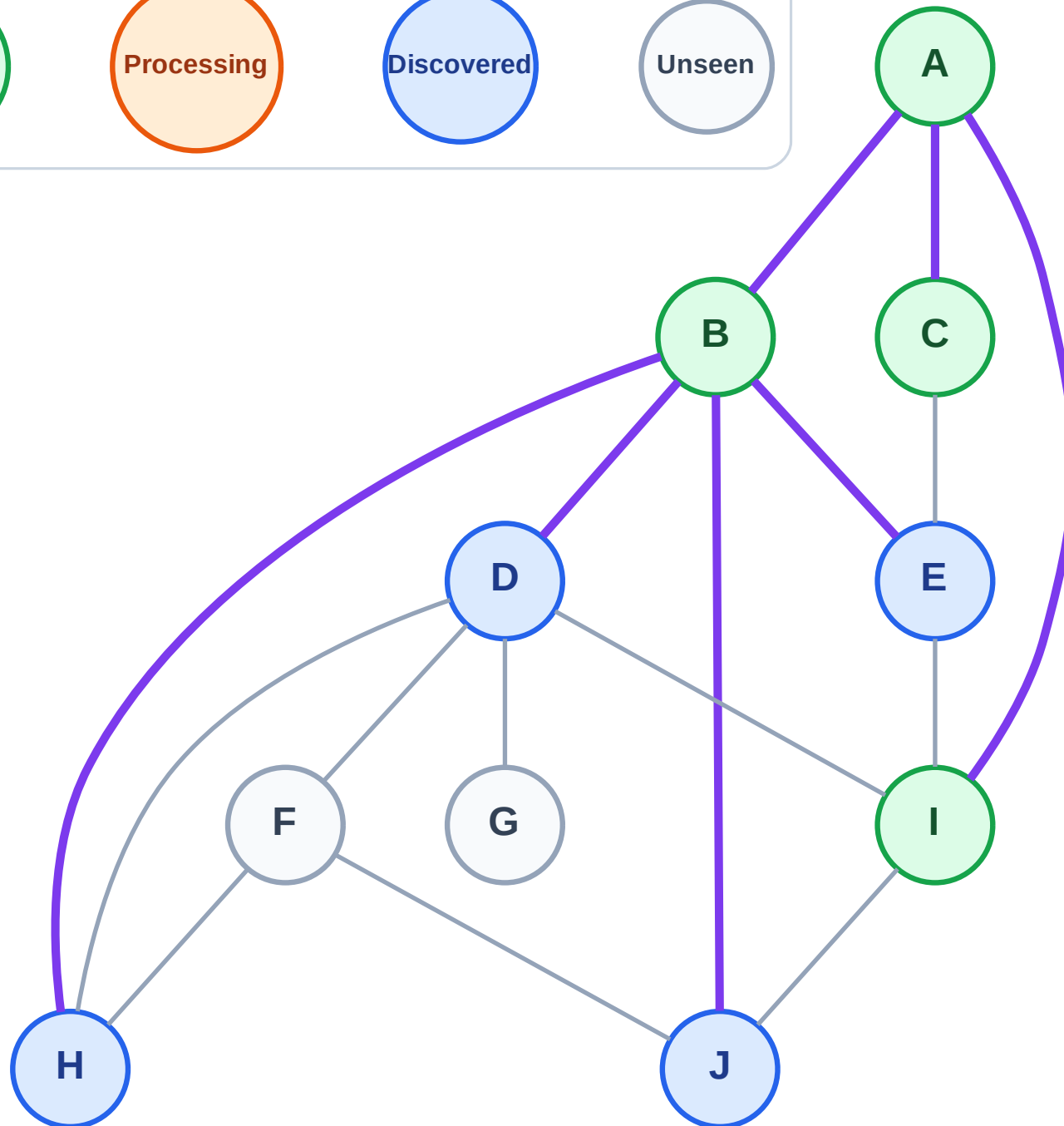
Step 9: No new neighbors from I

Legend



Queue: D → E → H → J

Visited: A, B, C, I



BFS Traversal

Step 10: Process node D

Legend

Complete

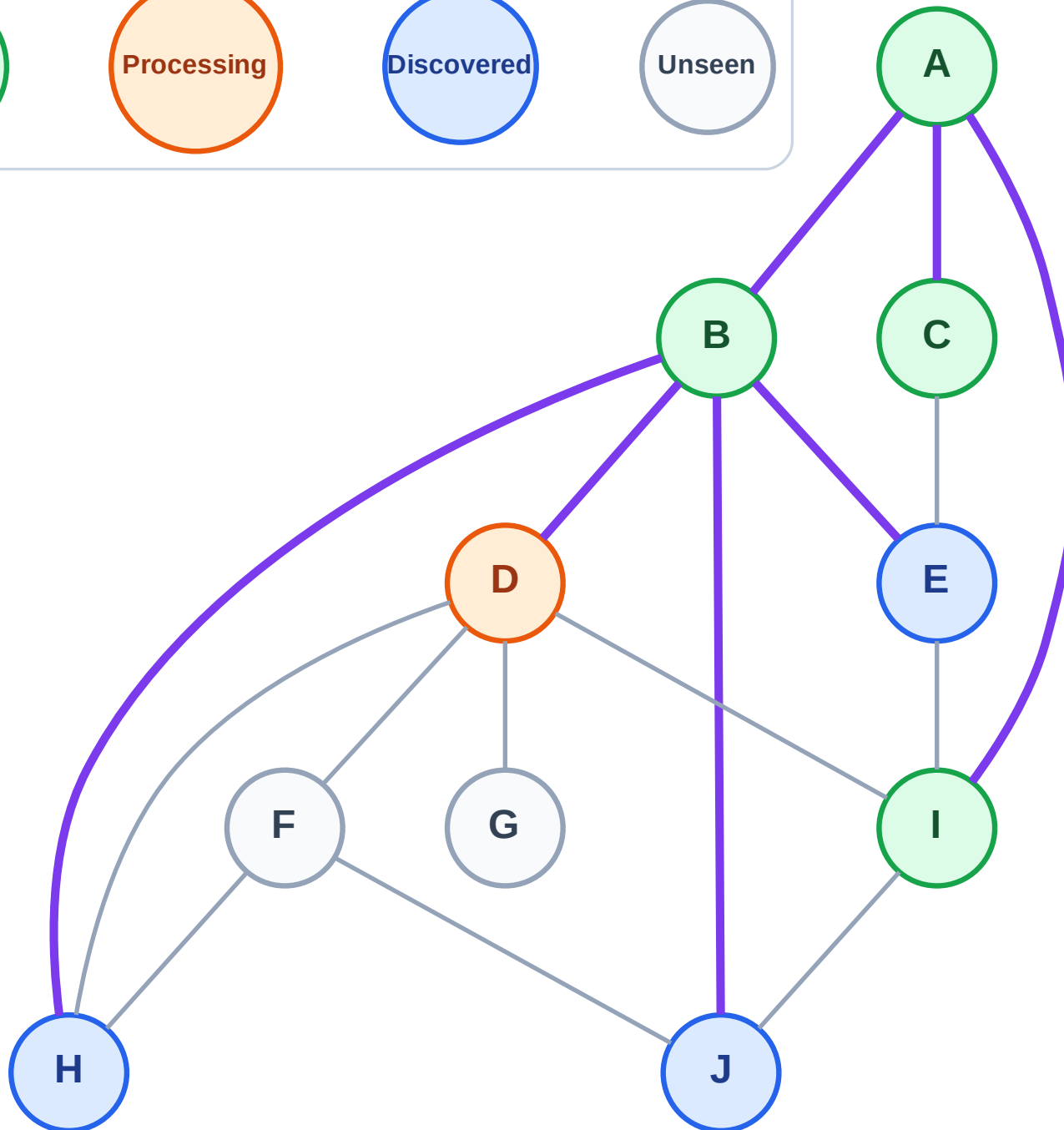
Processing

Discovered

Unseen

Queue: E → H → J

Visited: A, B, C, I



BFS Traversal

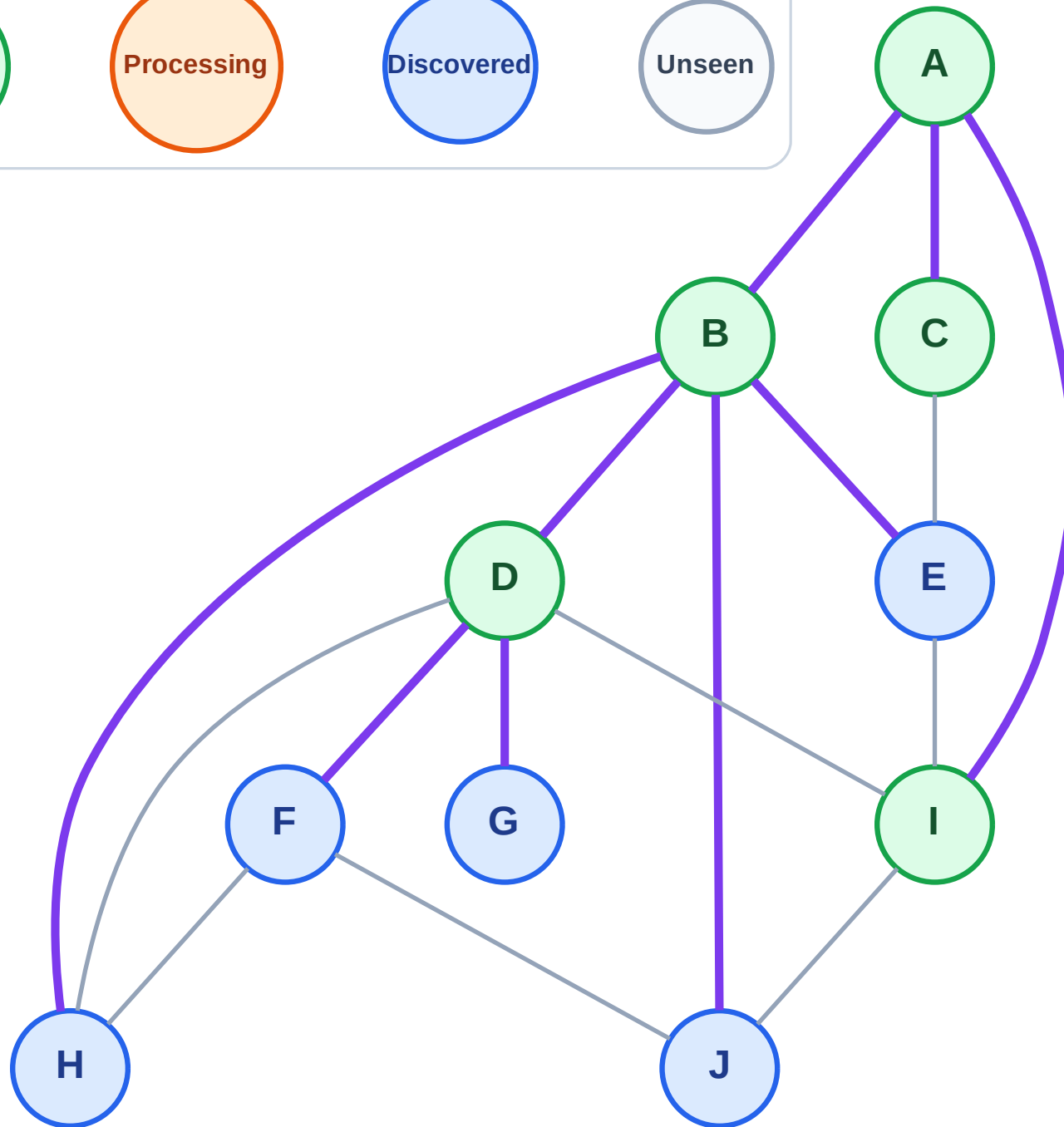
Step 11: Discover F, G from D

Legend



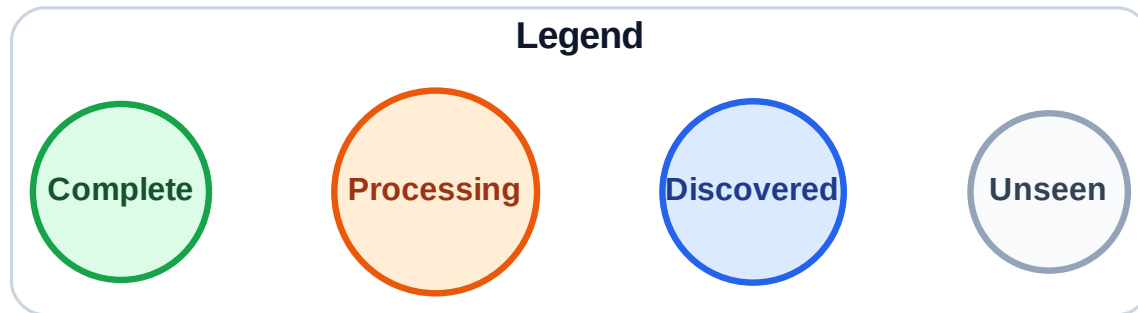
Queue: E → H → J → F → G

Visited: A, B, C, D, I



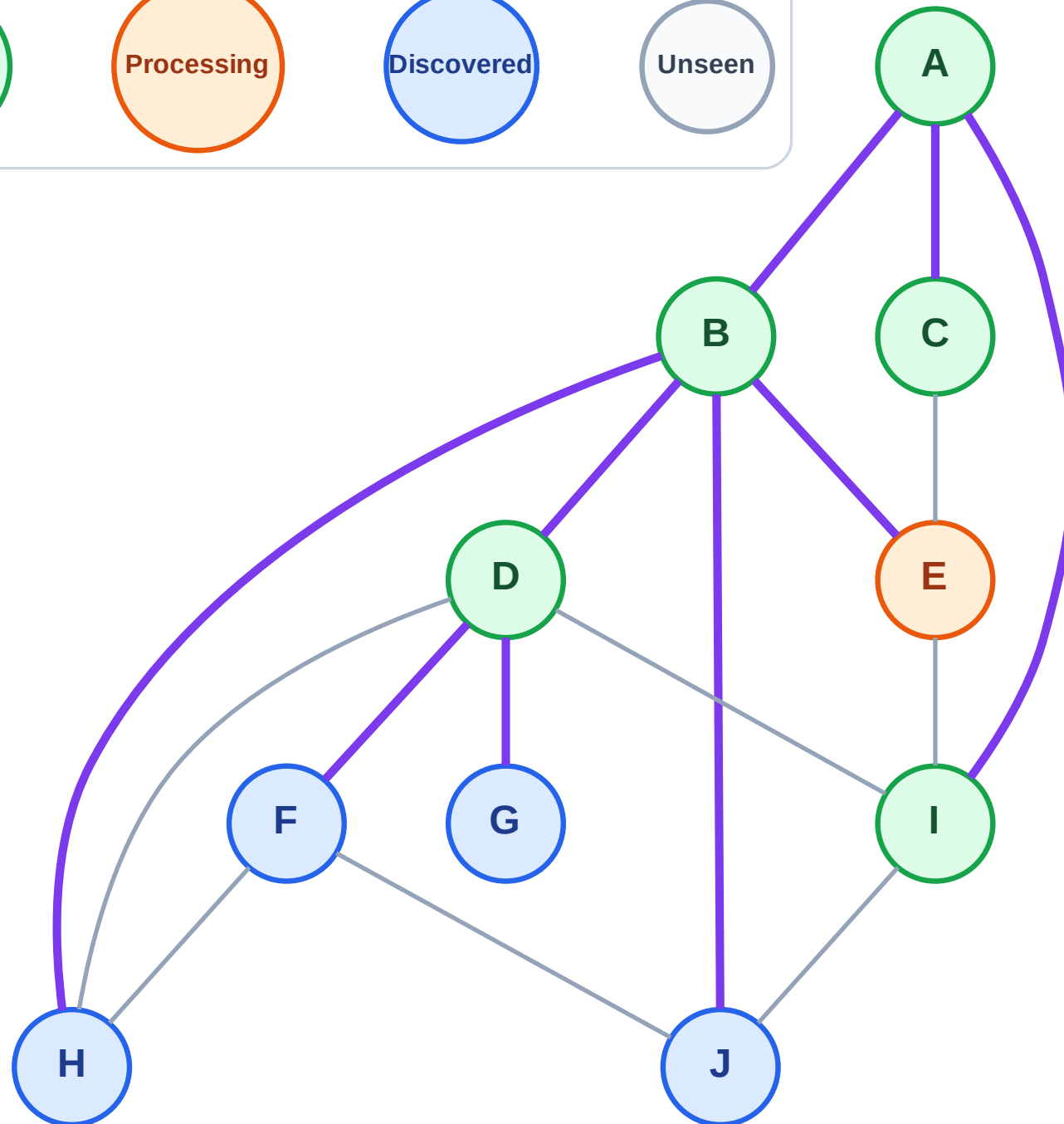
Step 12: Process node E

Step 12: Process node E



Queue: H → J → F → G

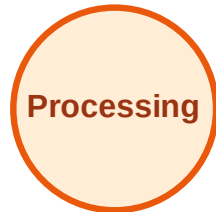
Visited: A, B, C, D, I



BFS Traversal

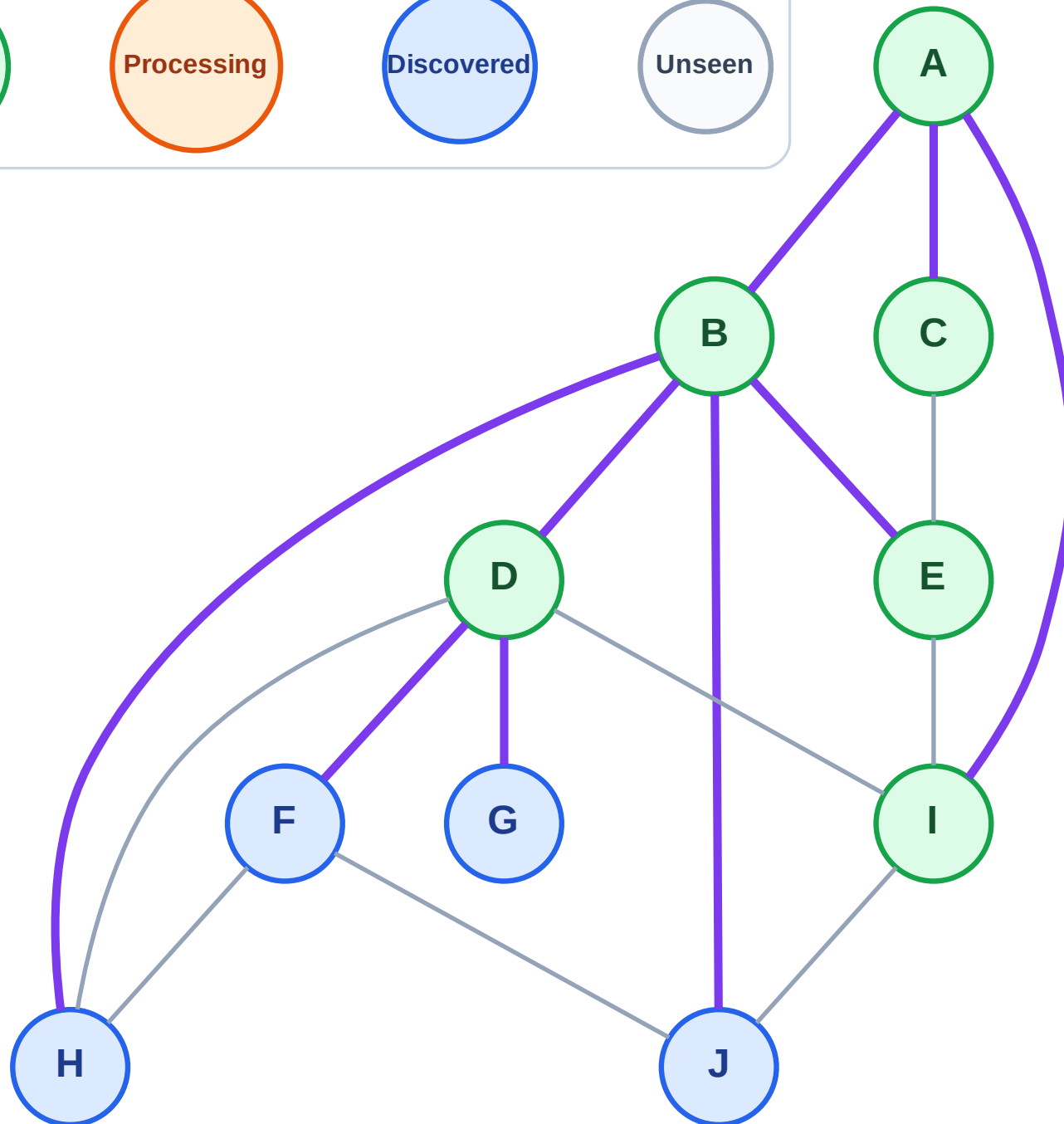
Step 13: No new neighbors from E

Legend



Queue: H → J → F → G

Visited: A, B, C, D, E, I



BFS Traversal

Step 14: Process node H

Legend

Complete

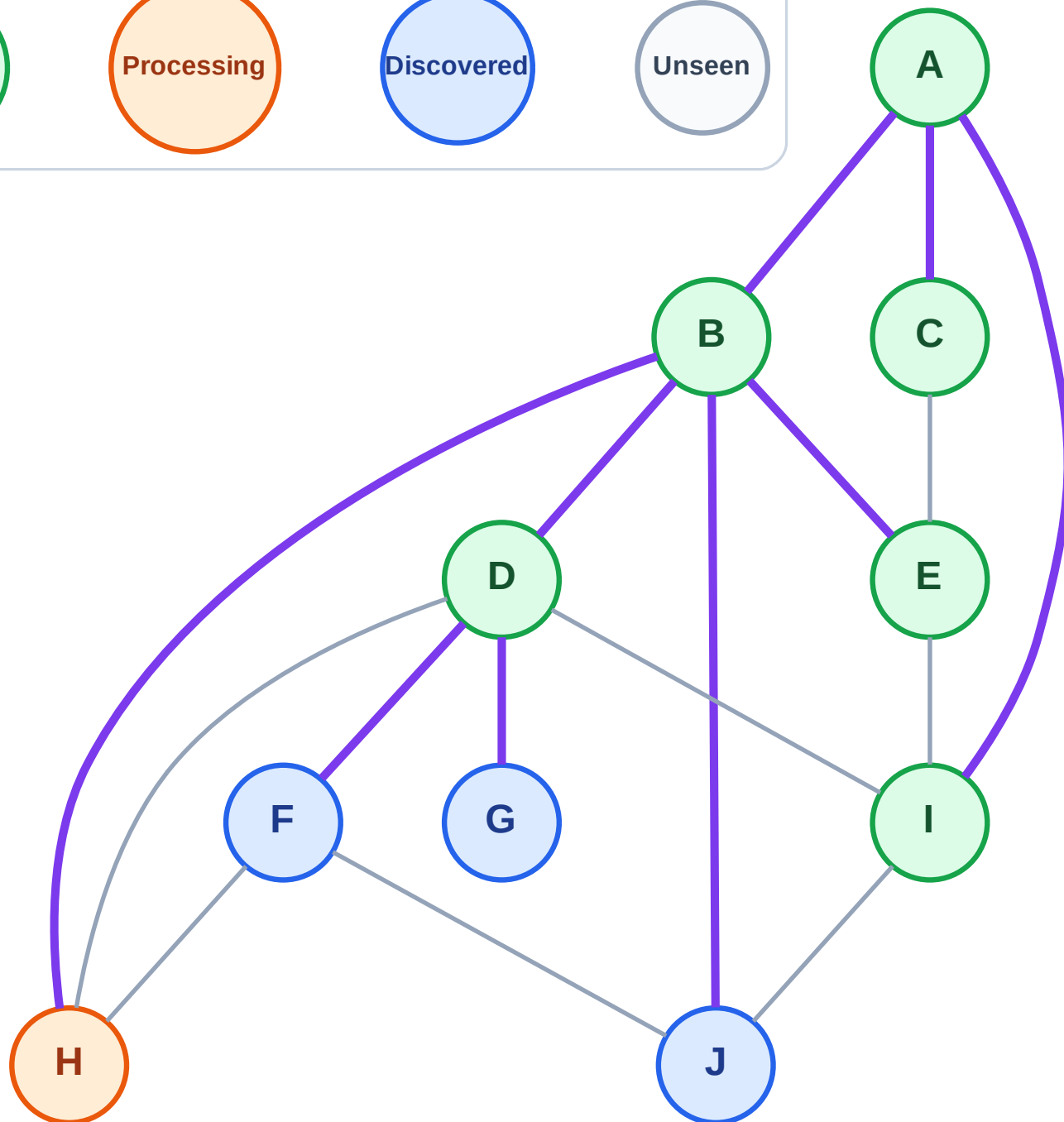
Processing

Discovered

Unseen

Queue: J → F → G

Visited: A, B, C, D, E, I



BFS Traversal

Step 15: No new neighbors from H

Legend

Complete

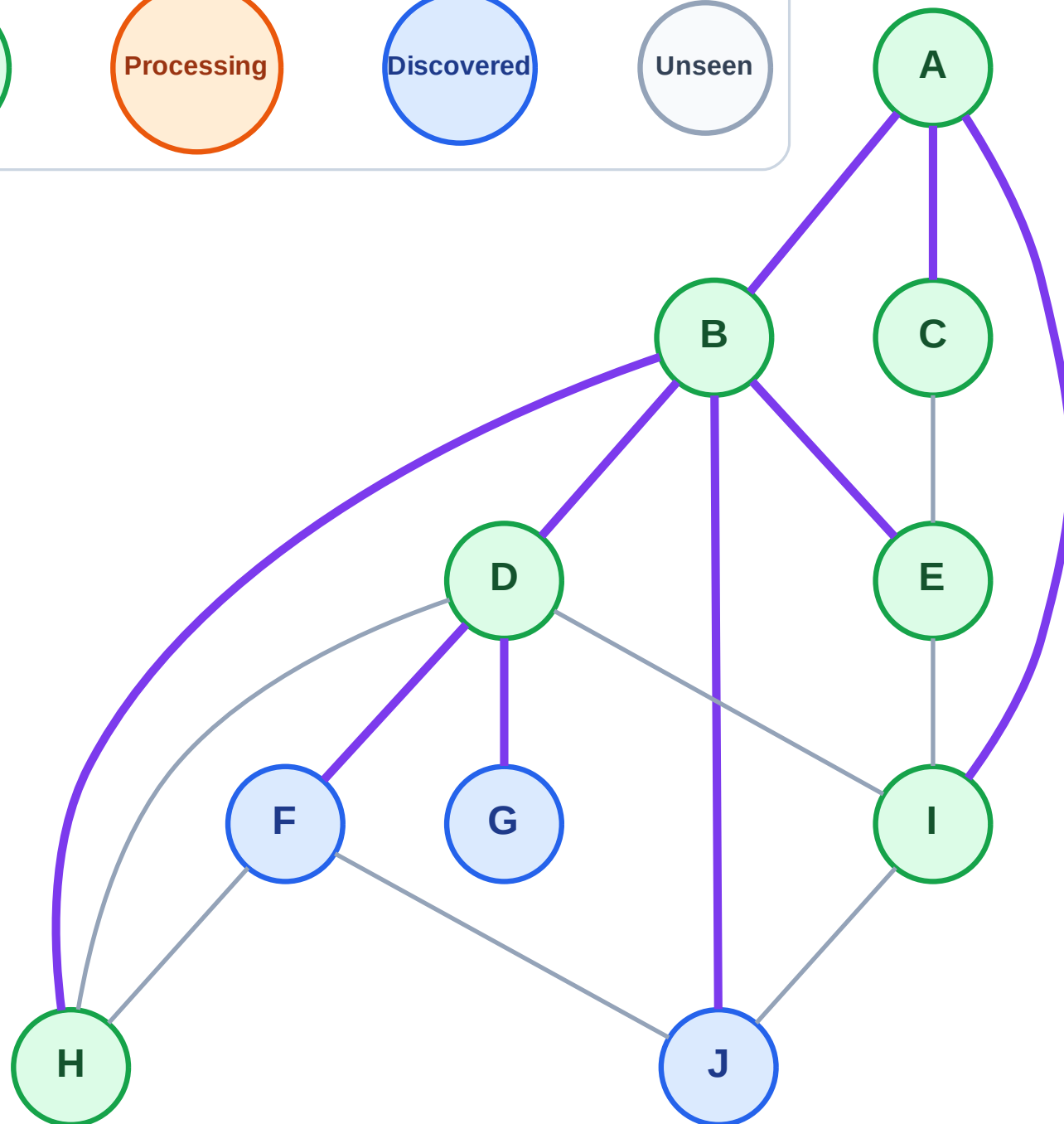
Processing

Discovered

Unseen

Queue: J → F → G

Visited: A, B, C, D, E, H, I



BFS Traversal

Step 16: Process node J

Legend

Complete

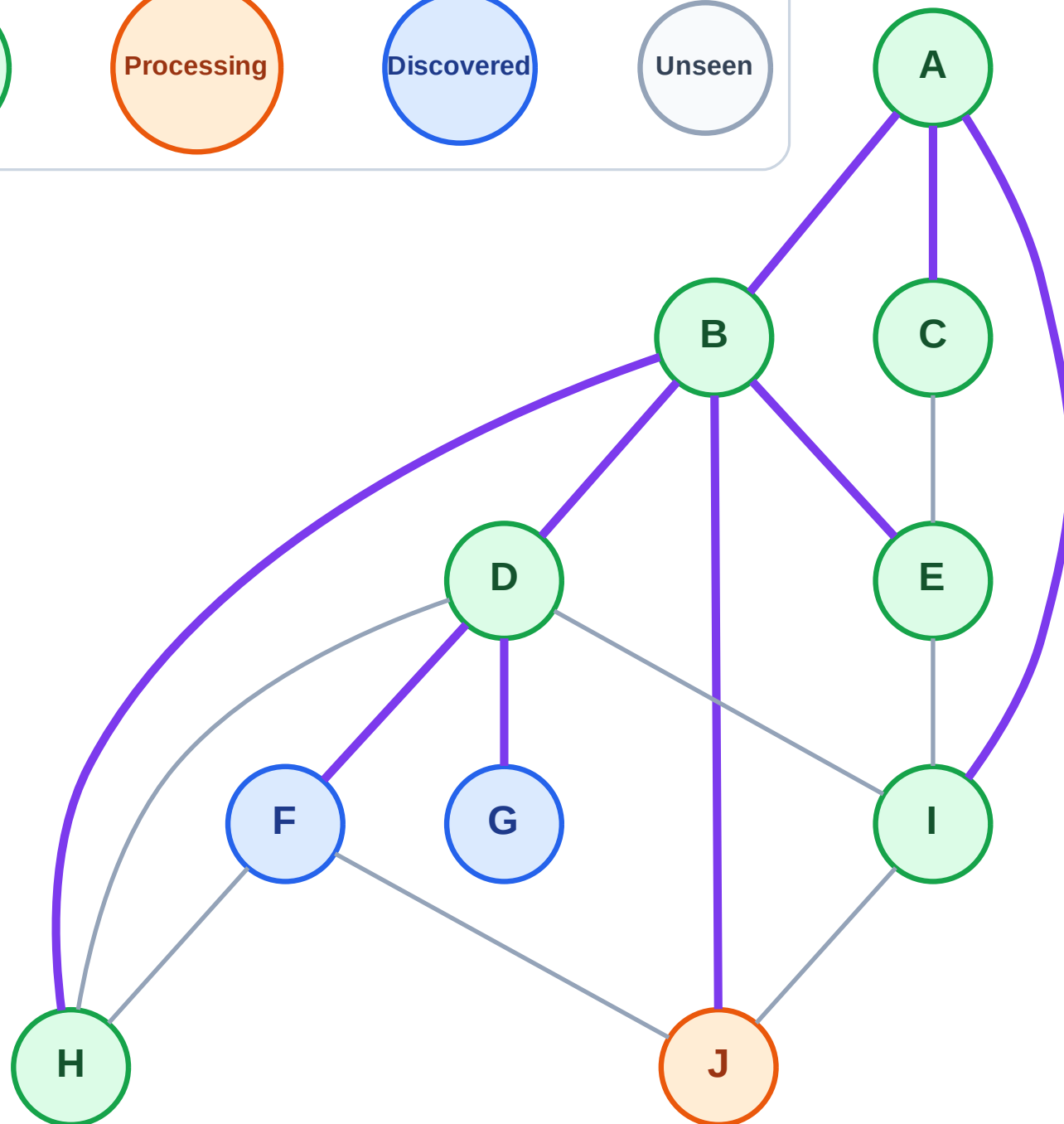
Processing

Discovered

Unseen

Queue: F → G

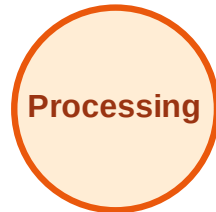
Visited: A, B, C, D, E, H, I



BFS Traversal

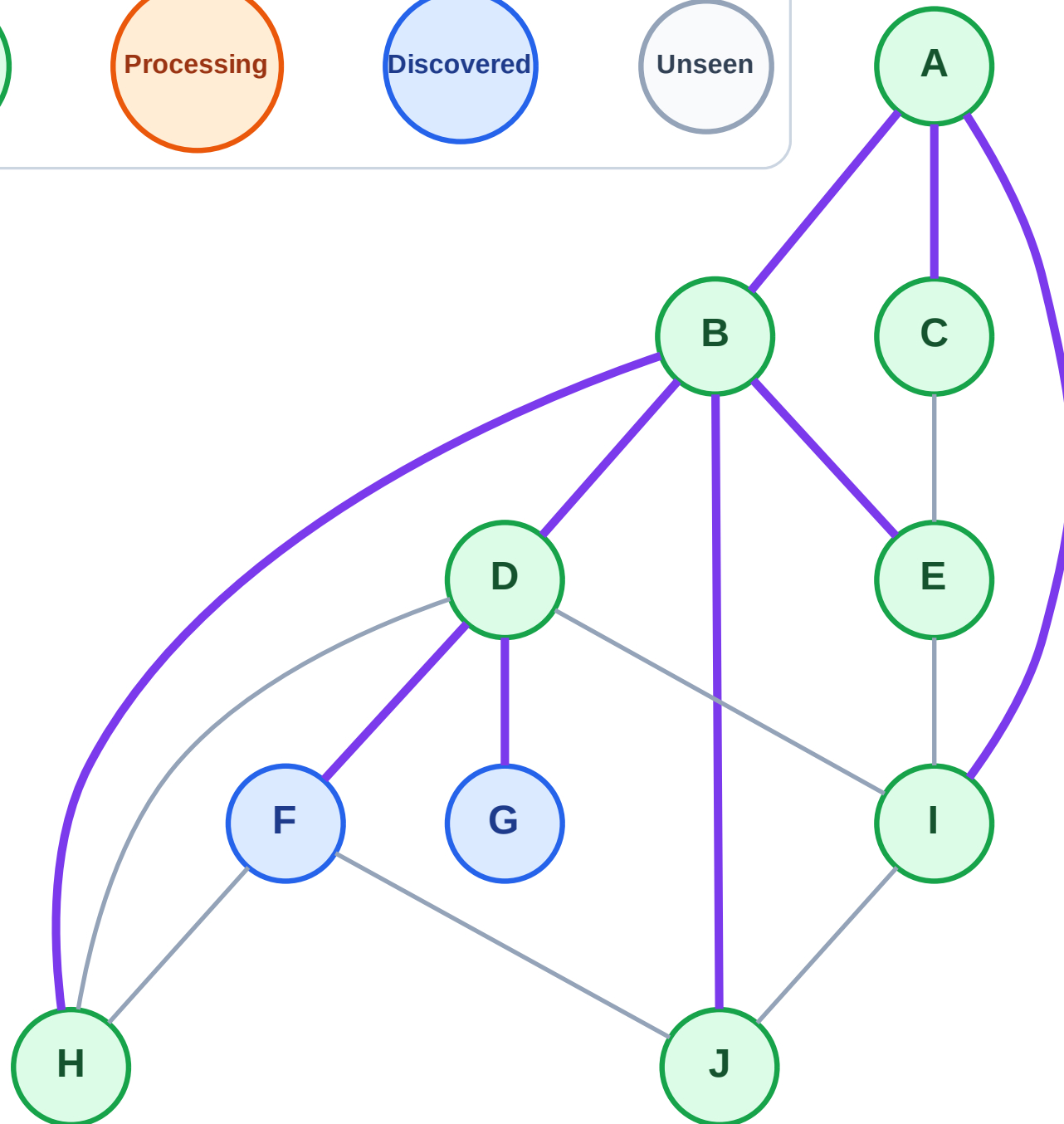
Step 17: No new neighbors from J

Legend



Queue: F → G

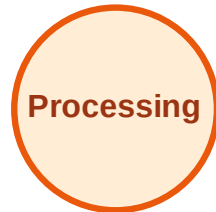
Visited: A, B, C, D, E, H, I, J



BFS Traversal

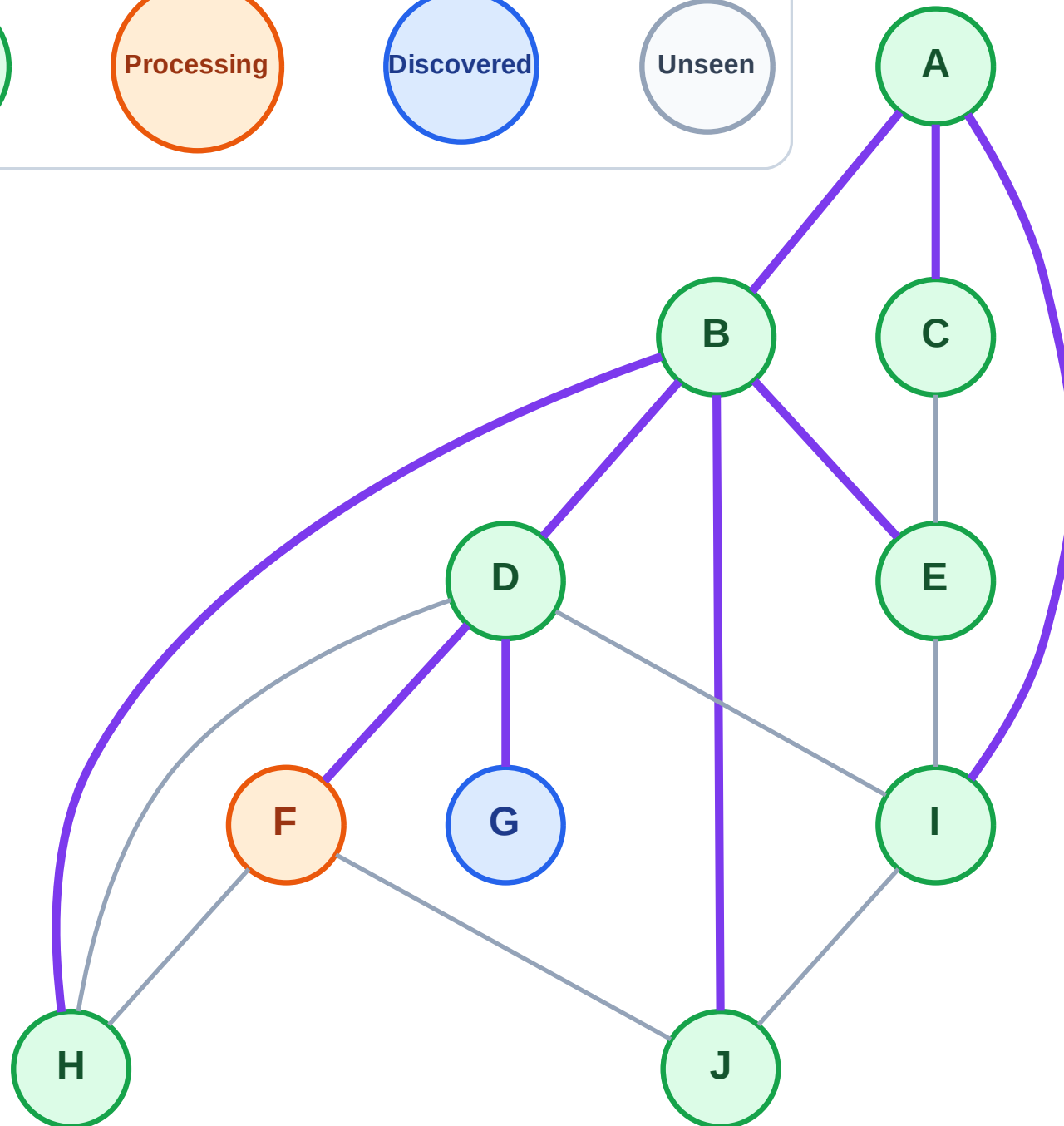
Step 18: Process node F

Legend



Queue: G

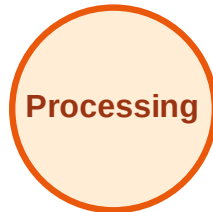
Visited: A, B, C, D, E, H, I, J



BFS Traversal

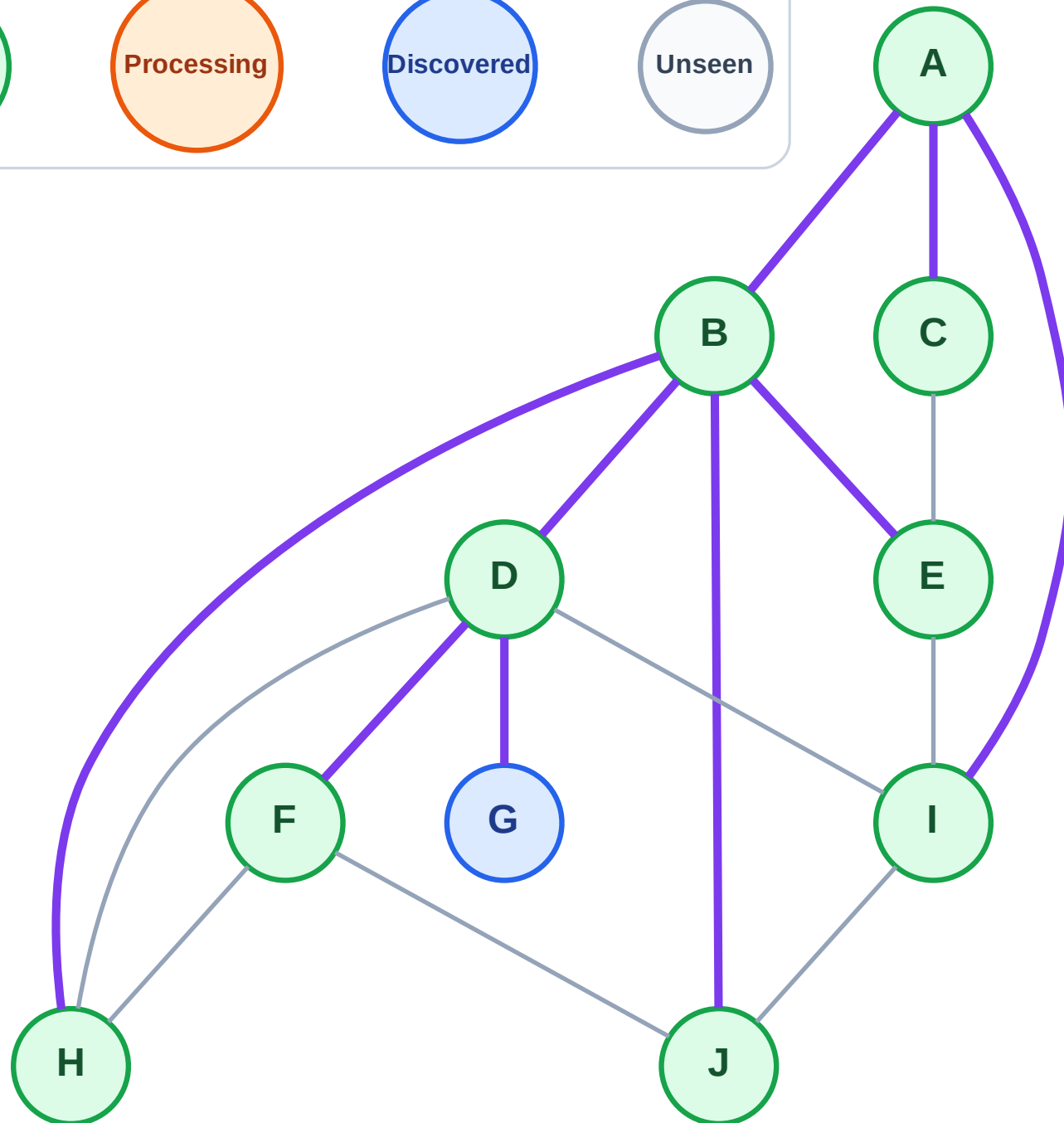
Step 19: No new neighbors from F

Legend



Queue: G

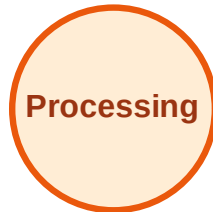
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

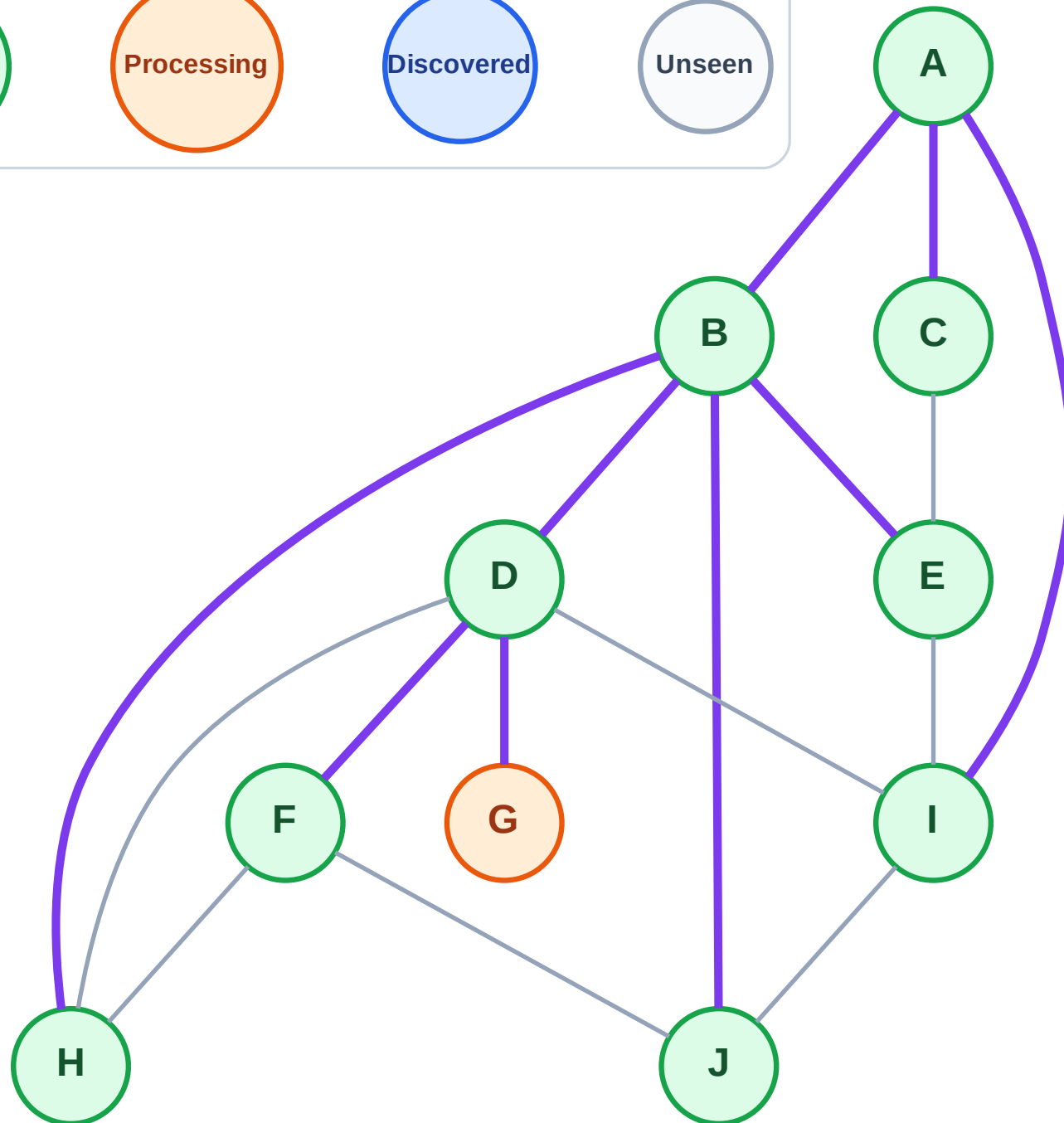
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

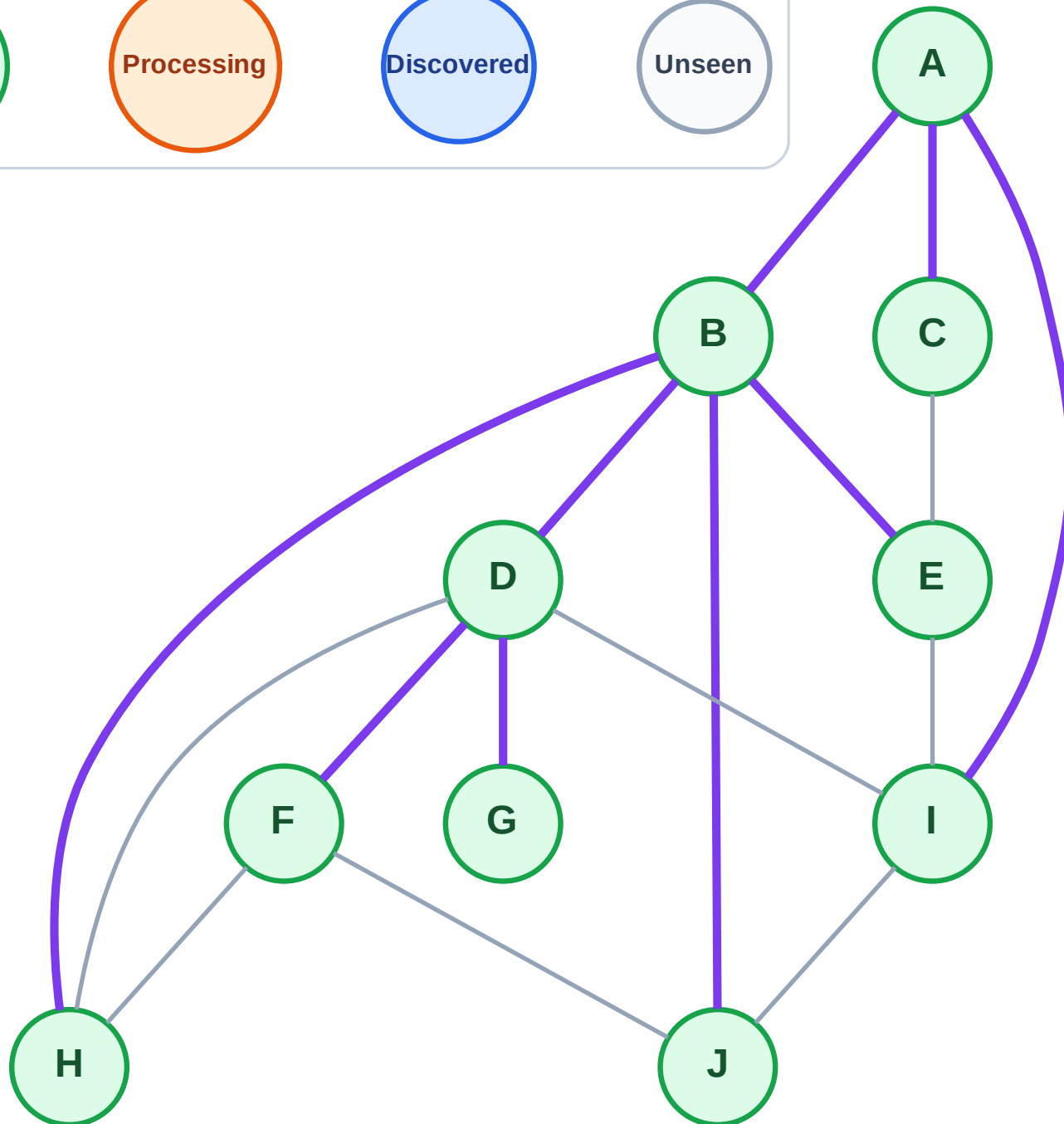
Step 21: No new neighbors from G

Legend



Queue: (empty)

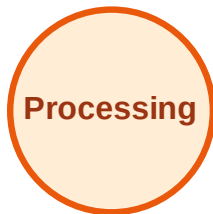
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

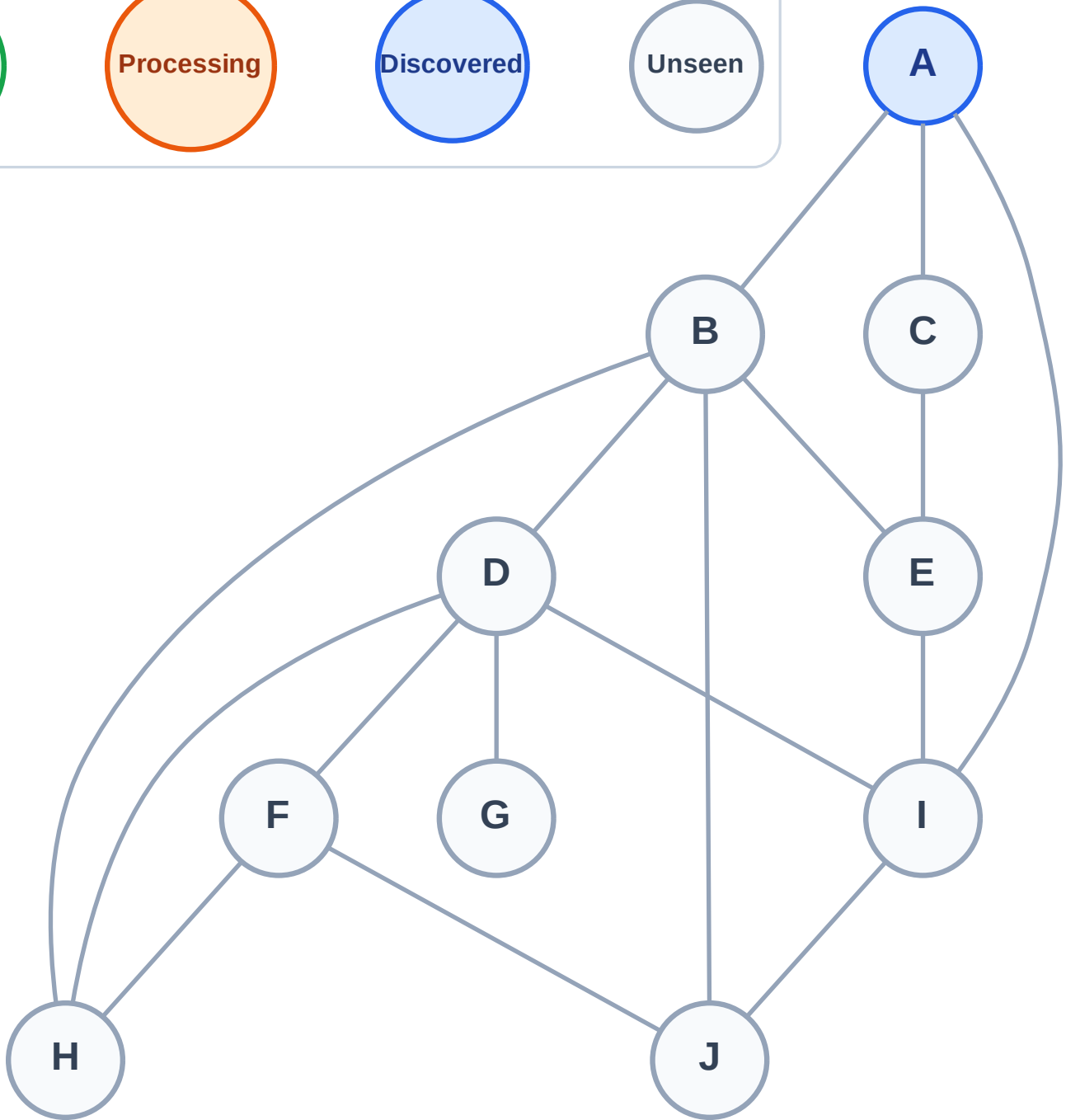
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

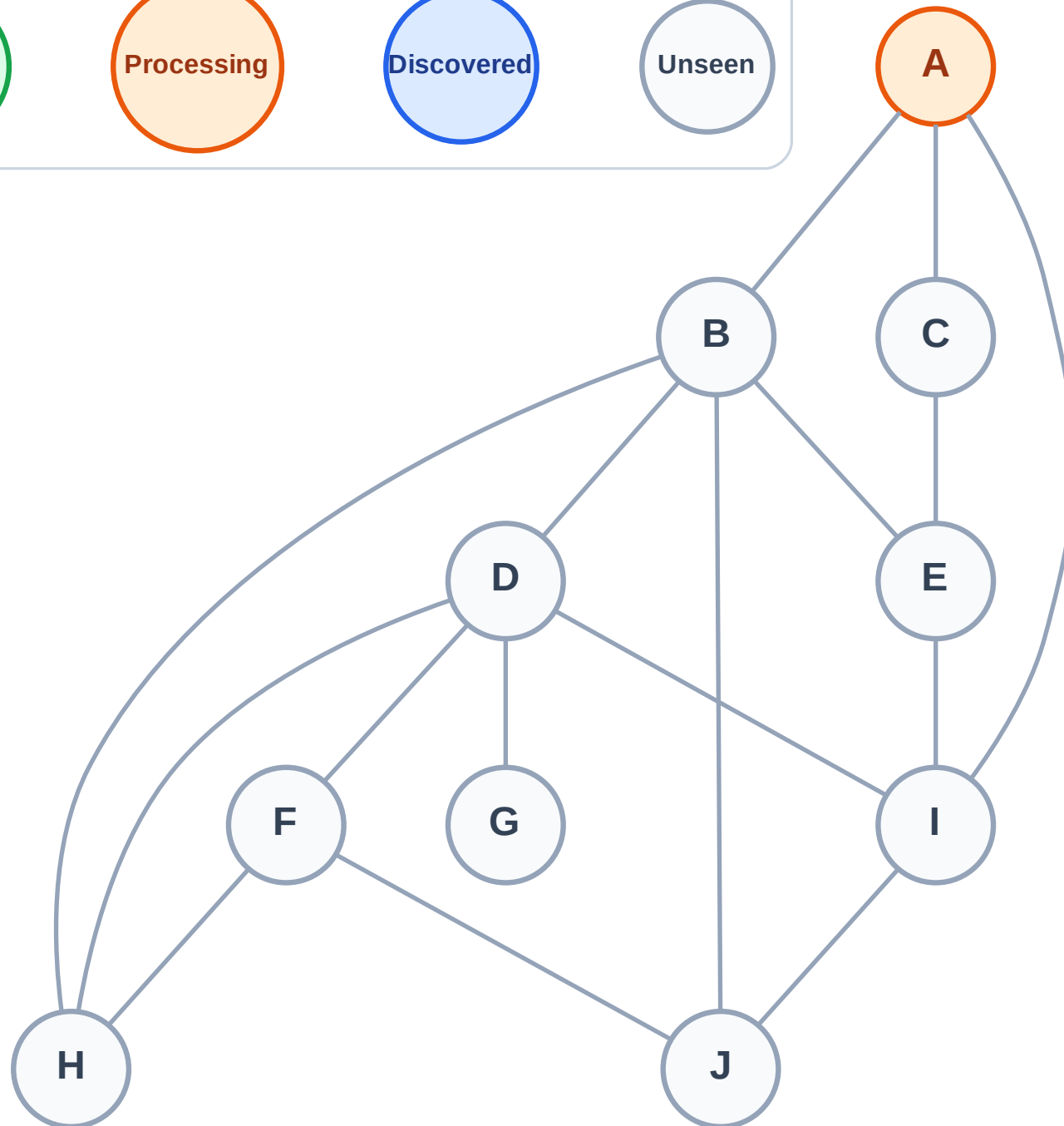
Processing

Discovered

Unseen

Stack (top at right): (empty)

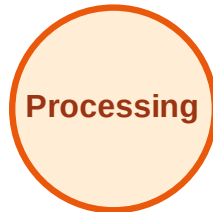
Visited: (none)



DFS Traversal

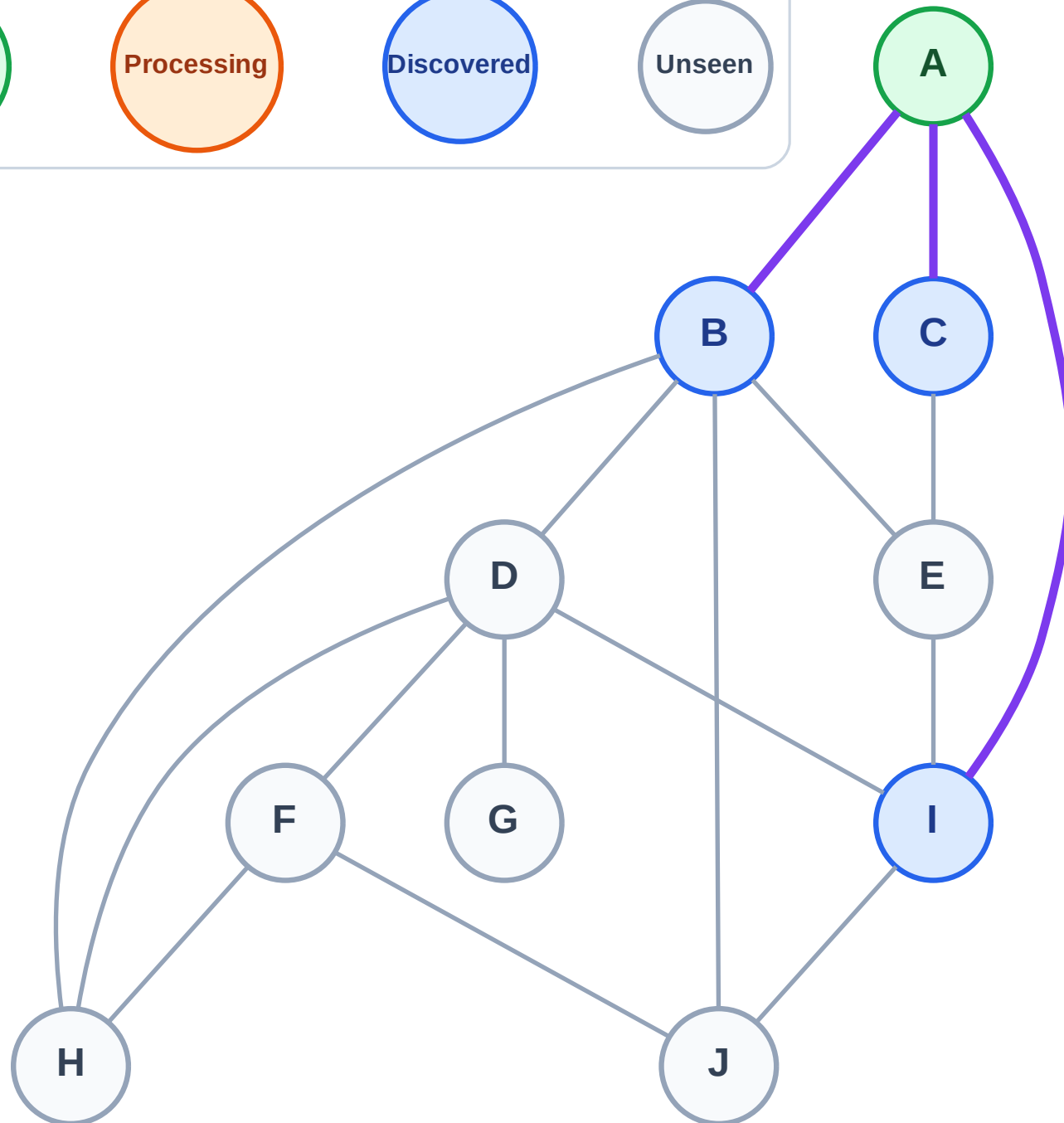
Step 3: Discover I, C, B from A

Legend



Stack (top at right): I | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

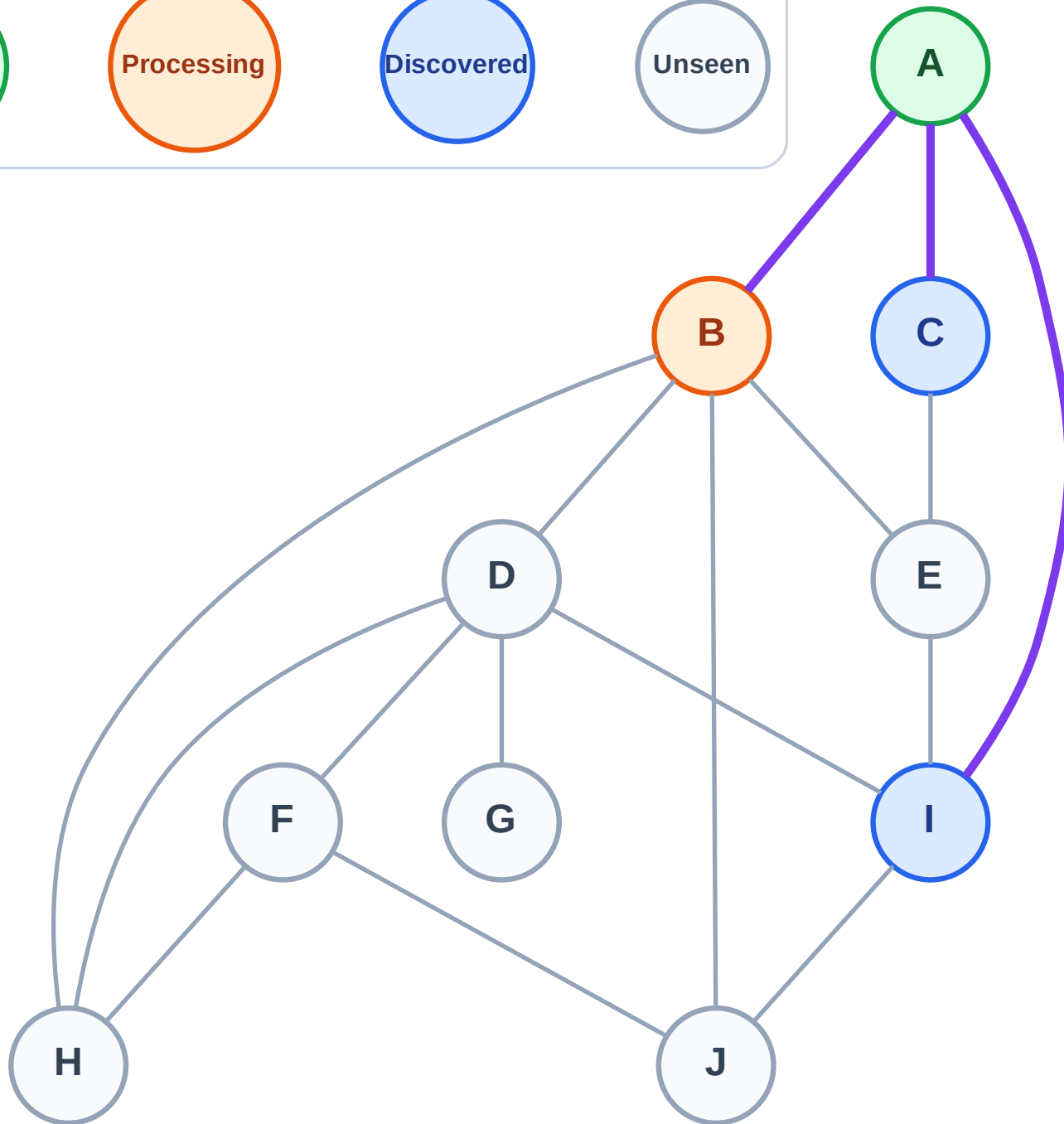
Processing

Discovered

Unseen

Stack (top at right): I | C

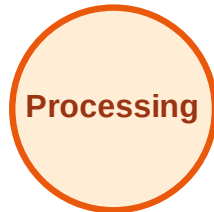
Visited: A



DFS Traversal

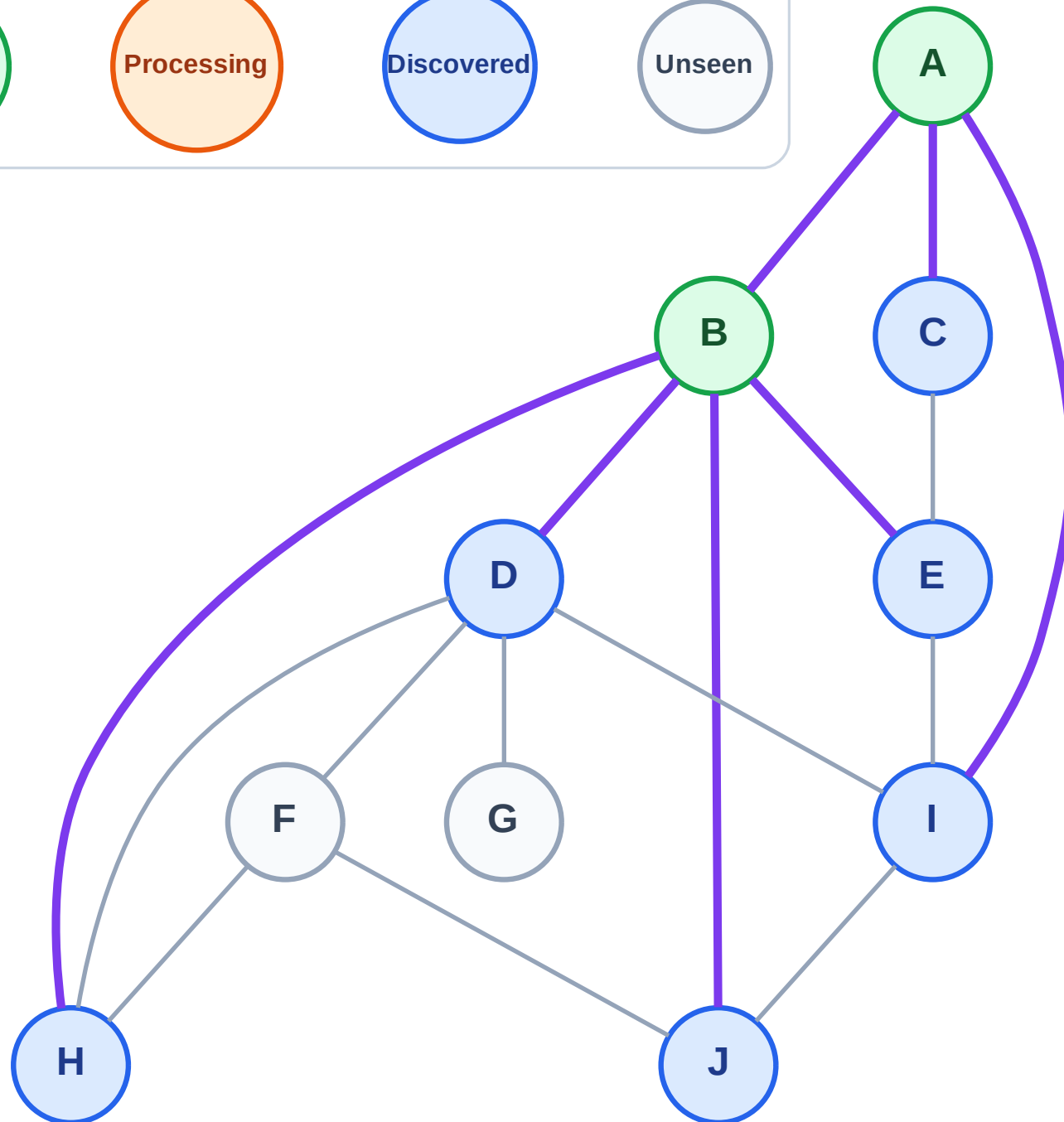
Step 5: Discover J, H, E, D from B

Legend



Stack (top at right): I | C | J | H | E | D

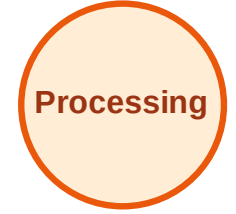
Visited: A, B



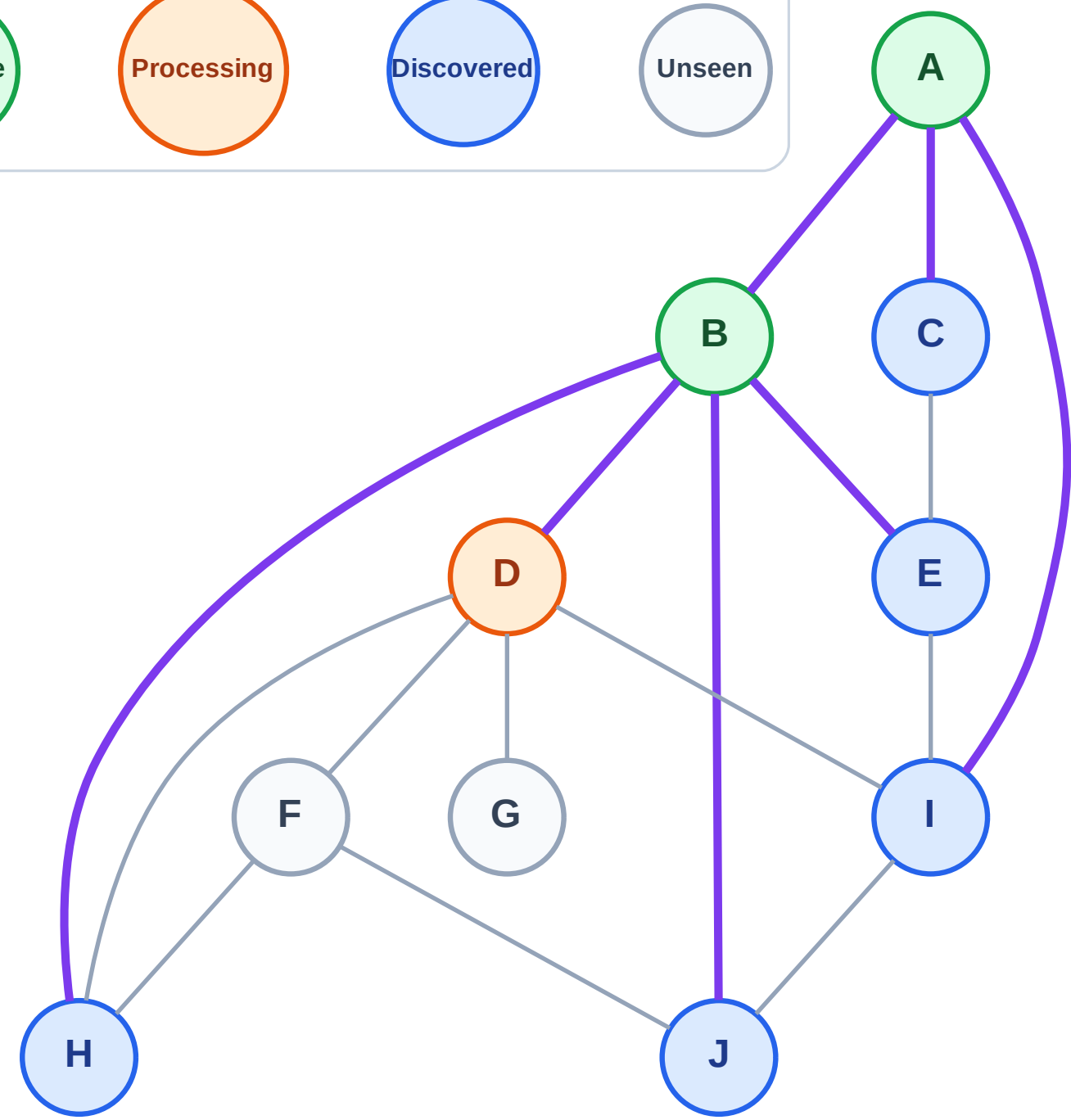
DFS Traversal

Step 6: Process node D

Legend



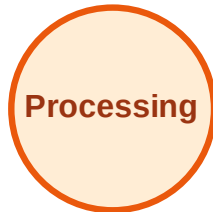
Stack (top at right): I | C | J | H | E
Visited: A, B



DFS Traversal

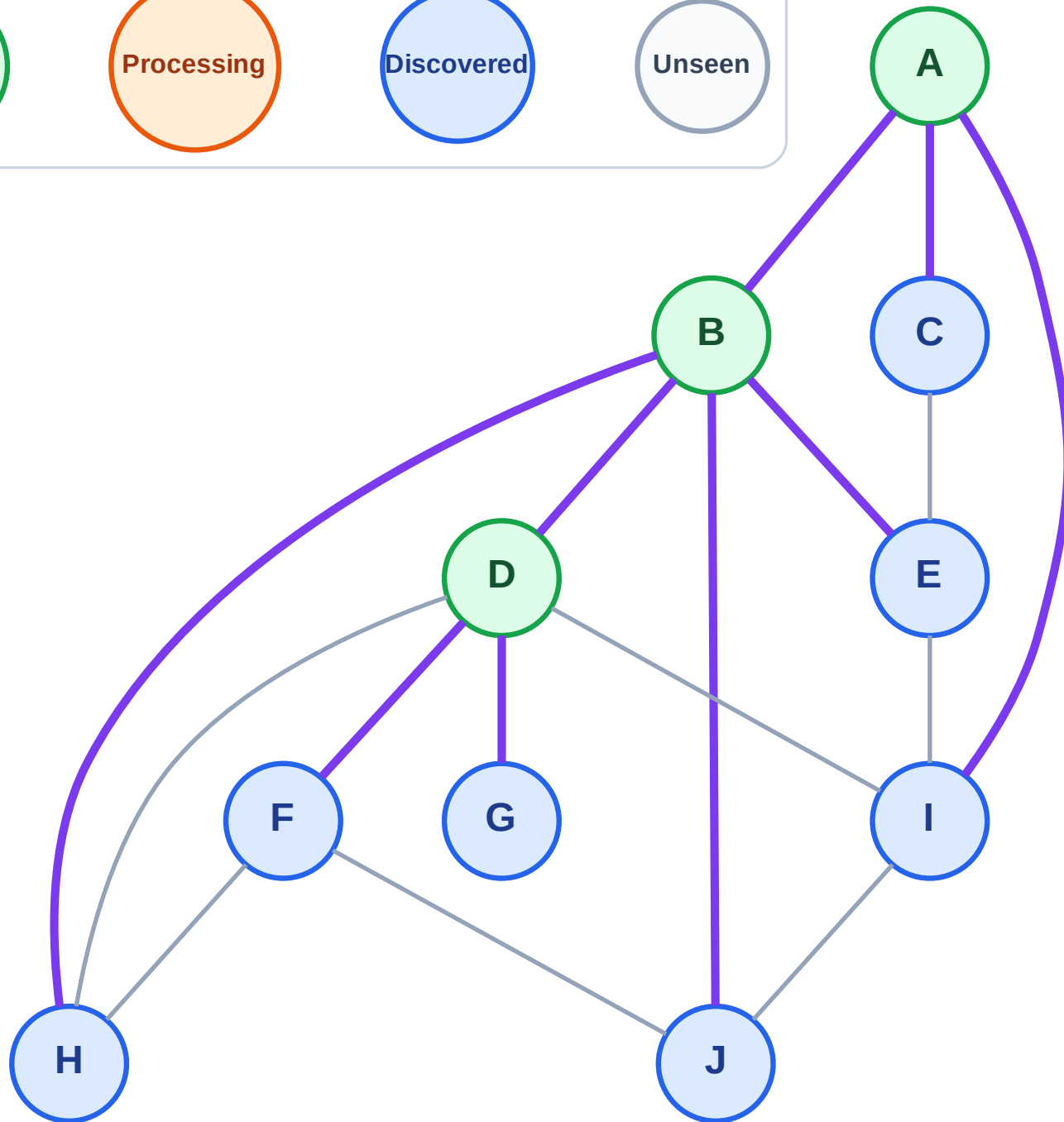
Step 7: Discover G, F from D

Legend

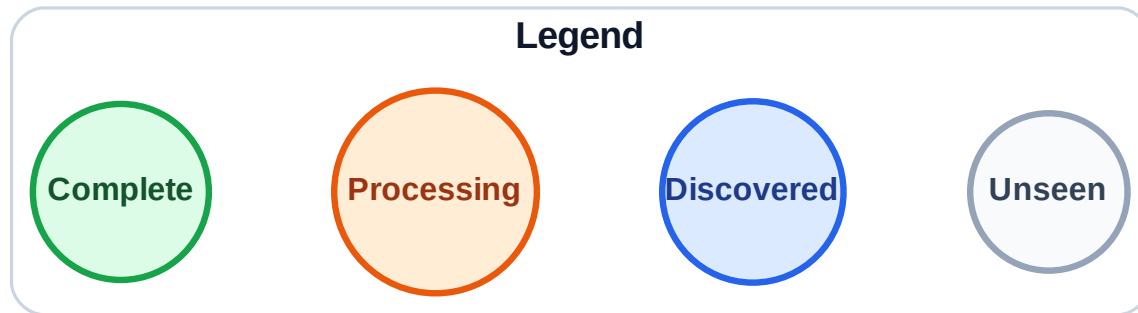


Stack (top at right): I | C | J | H | E | G | F

Visited: A, B, D

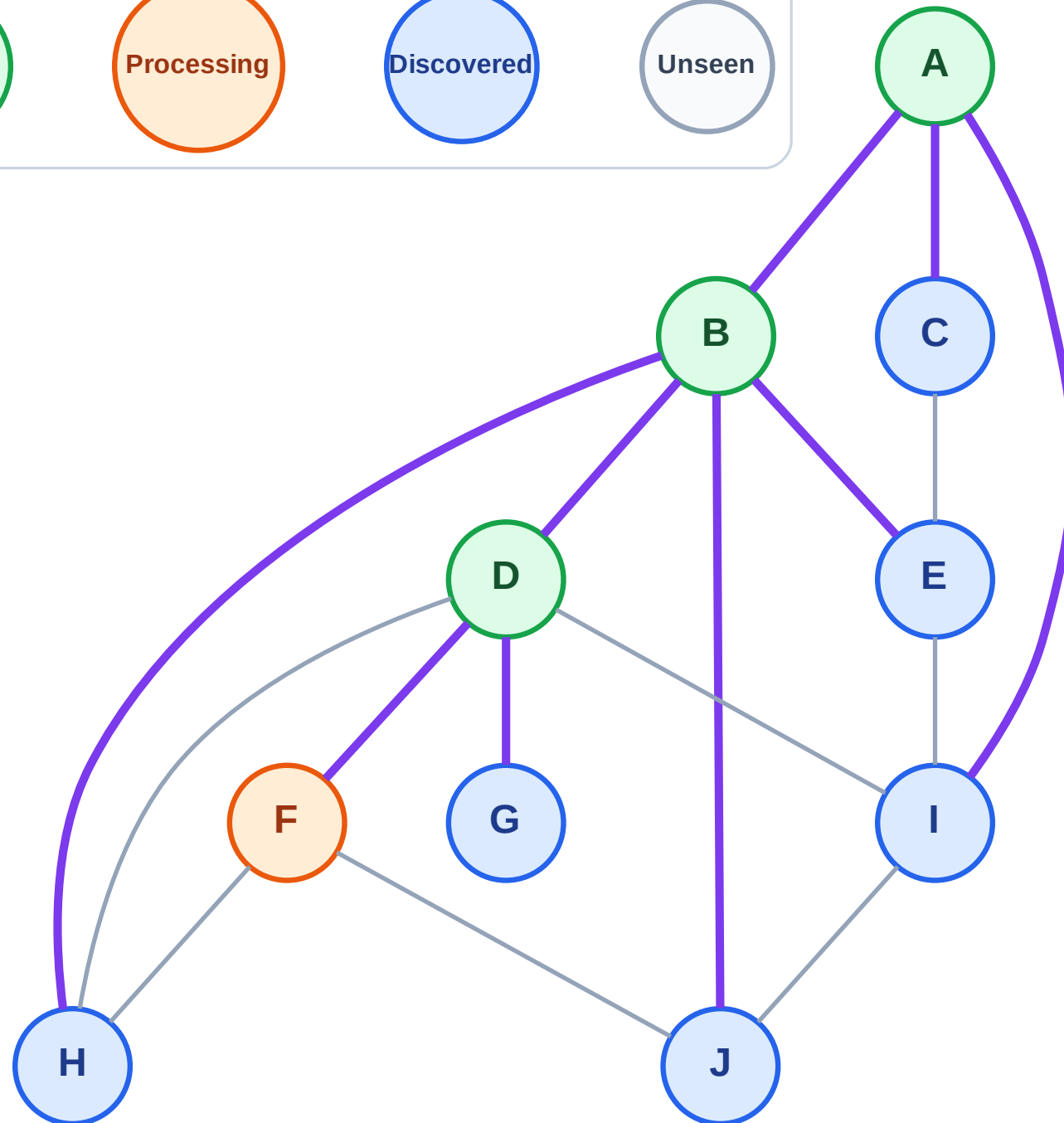


Step 8: Process node F



Stack (top at right): I | C | J | H | E | G

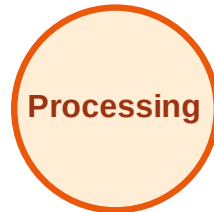
Visited: A, B, D



DFS Traversal

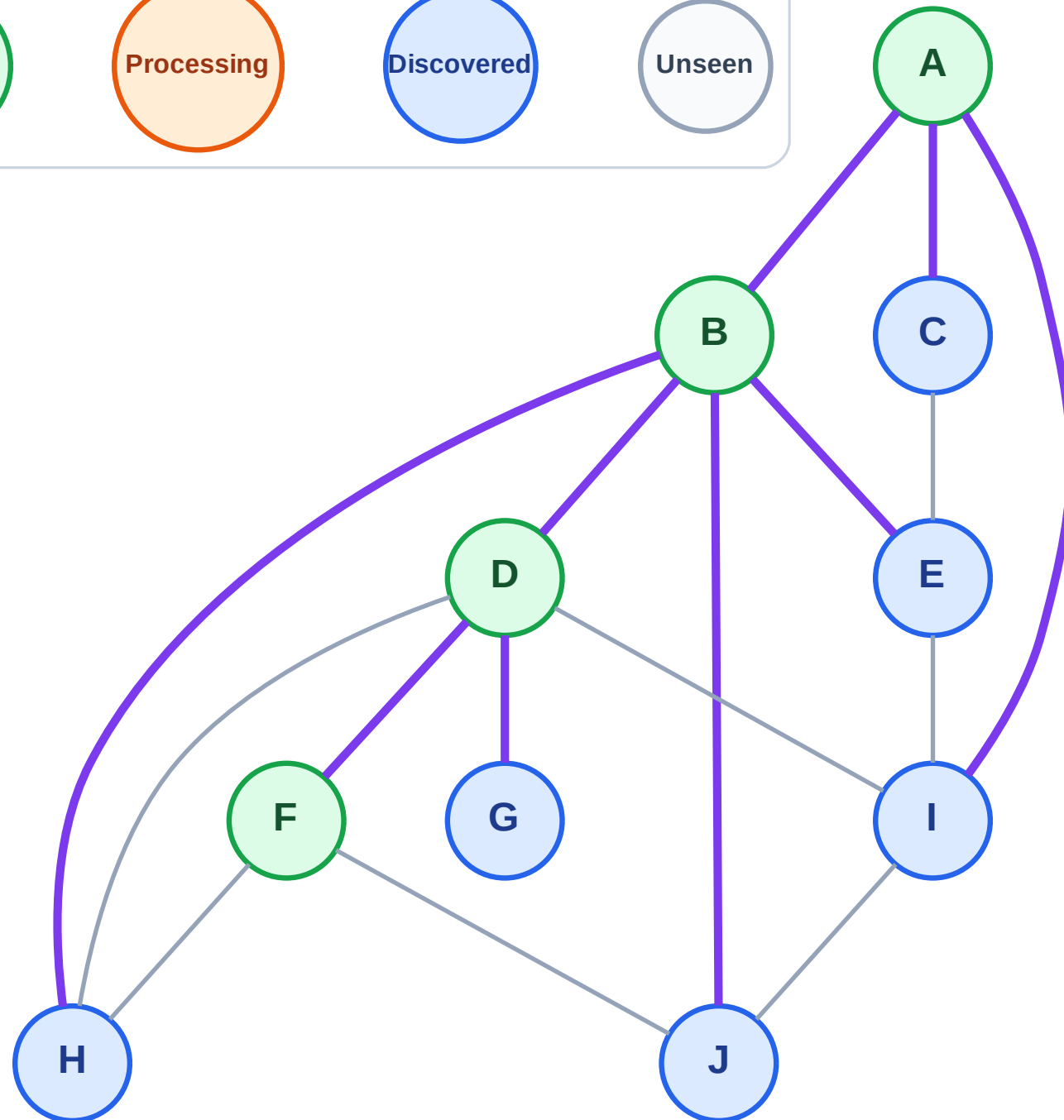
Step 9: No new neighbors from F

Legend



Stack (top at right): I | C | J | H | E | G

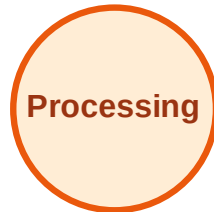
Visited: A, B, D, F



DFS Traversal

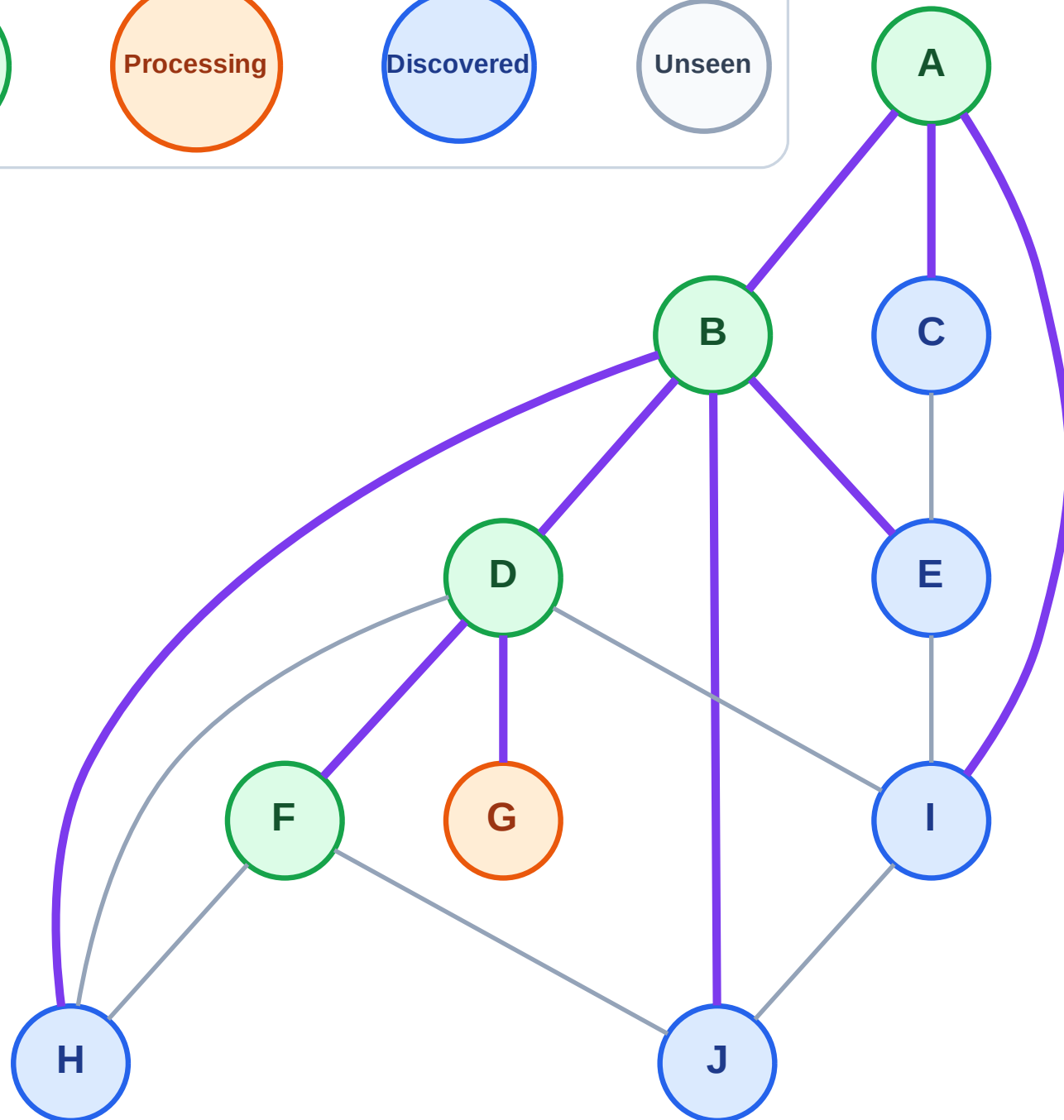
Step 10: Process node G

Legend



Stack (top at right): I | C | J | H | E

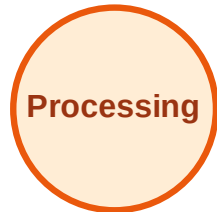
Visited: A, B, D, F



DFS Traversal

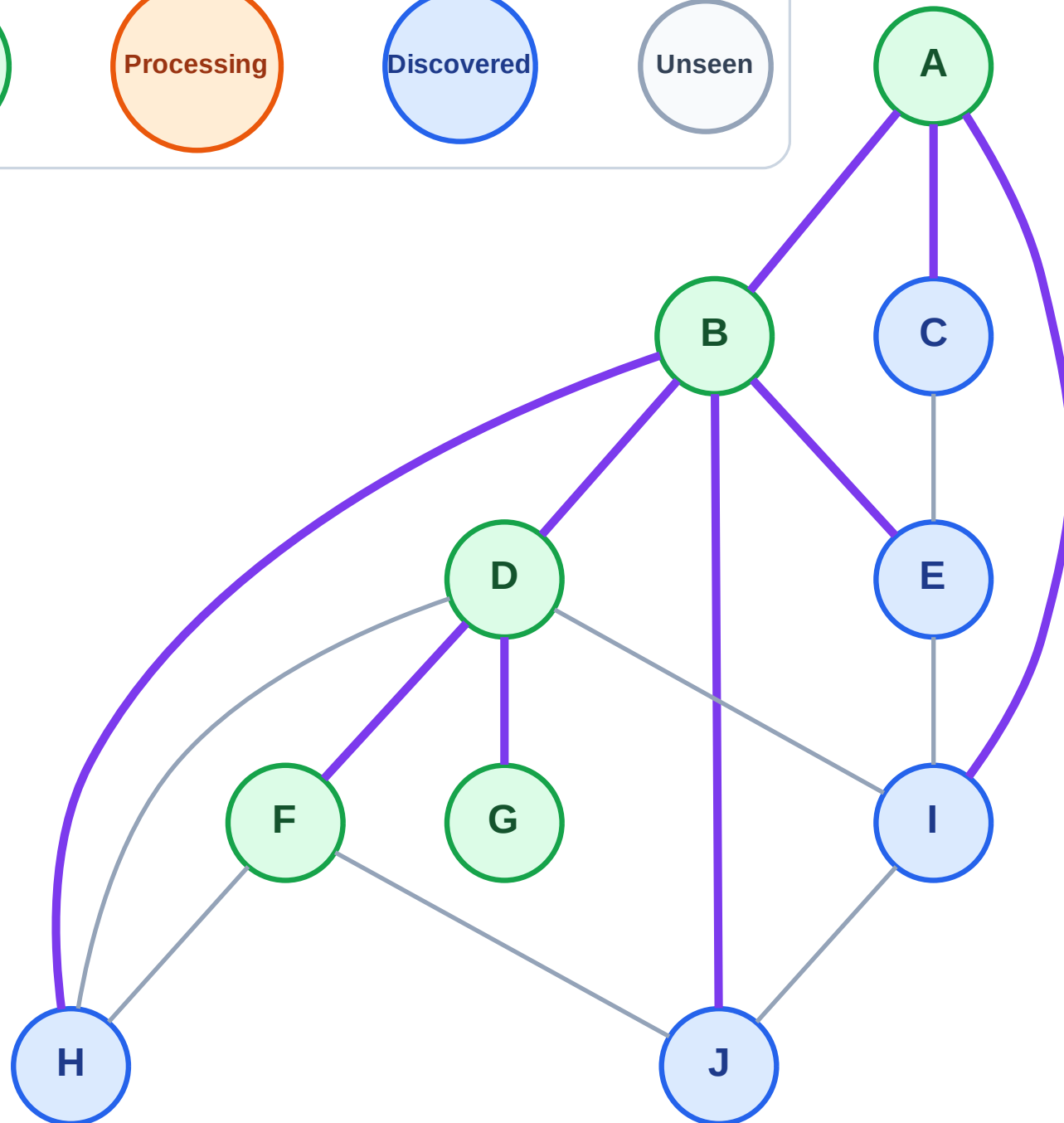
Step 11: No new neighbors from G

Legend



Stack (top at right): I | C | J | H | E

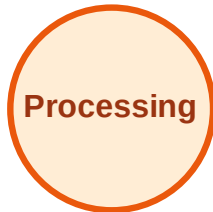
Visited: A, B, D, F, G



DFS Traversal

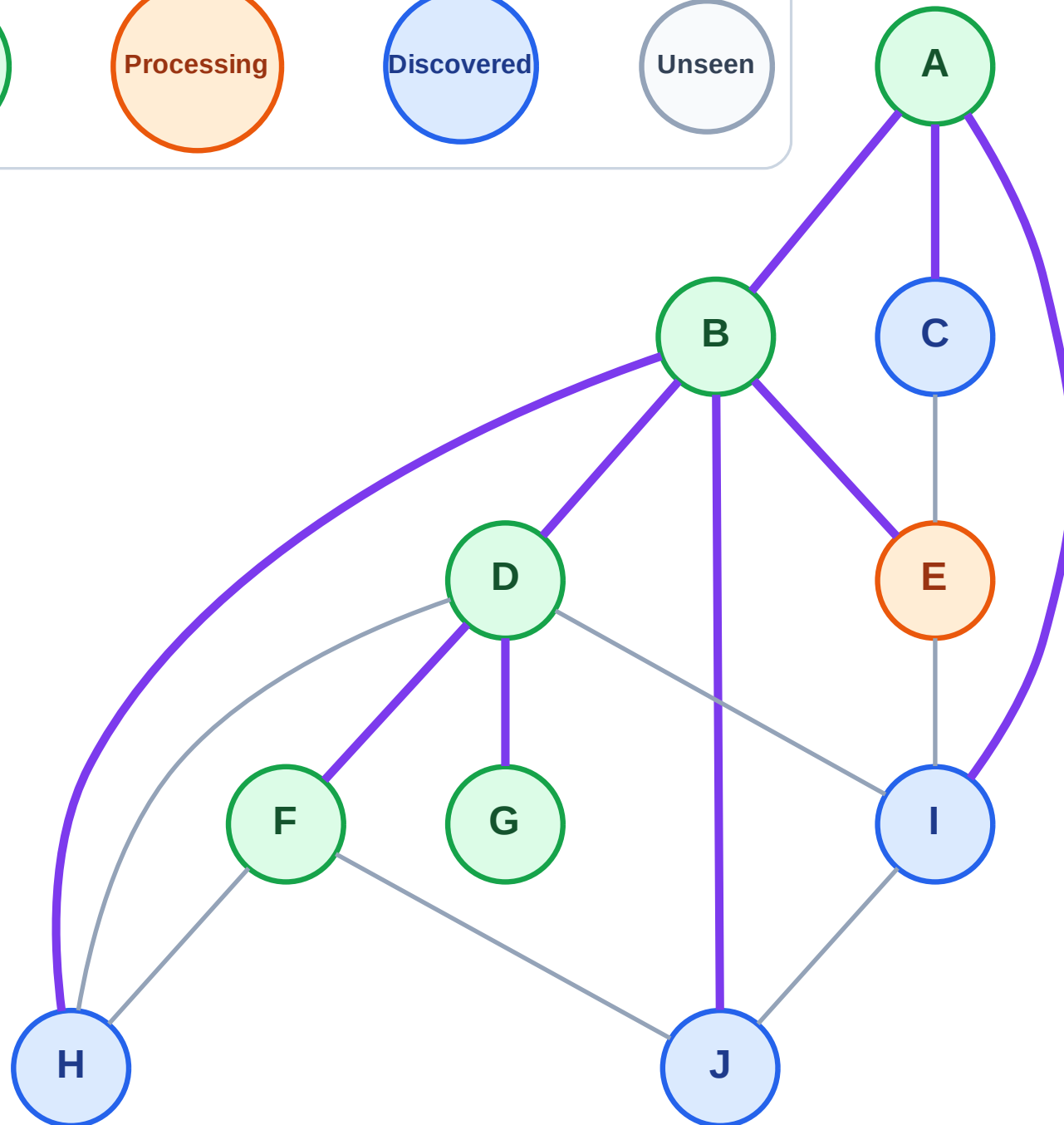
Step 12: Process node E

Legend



Stack (top at right): I | C | J | H

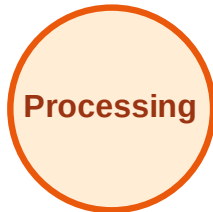
Visited: A, B, D, F, G



DFS Traversal

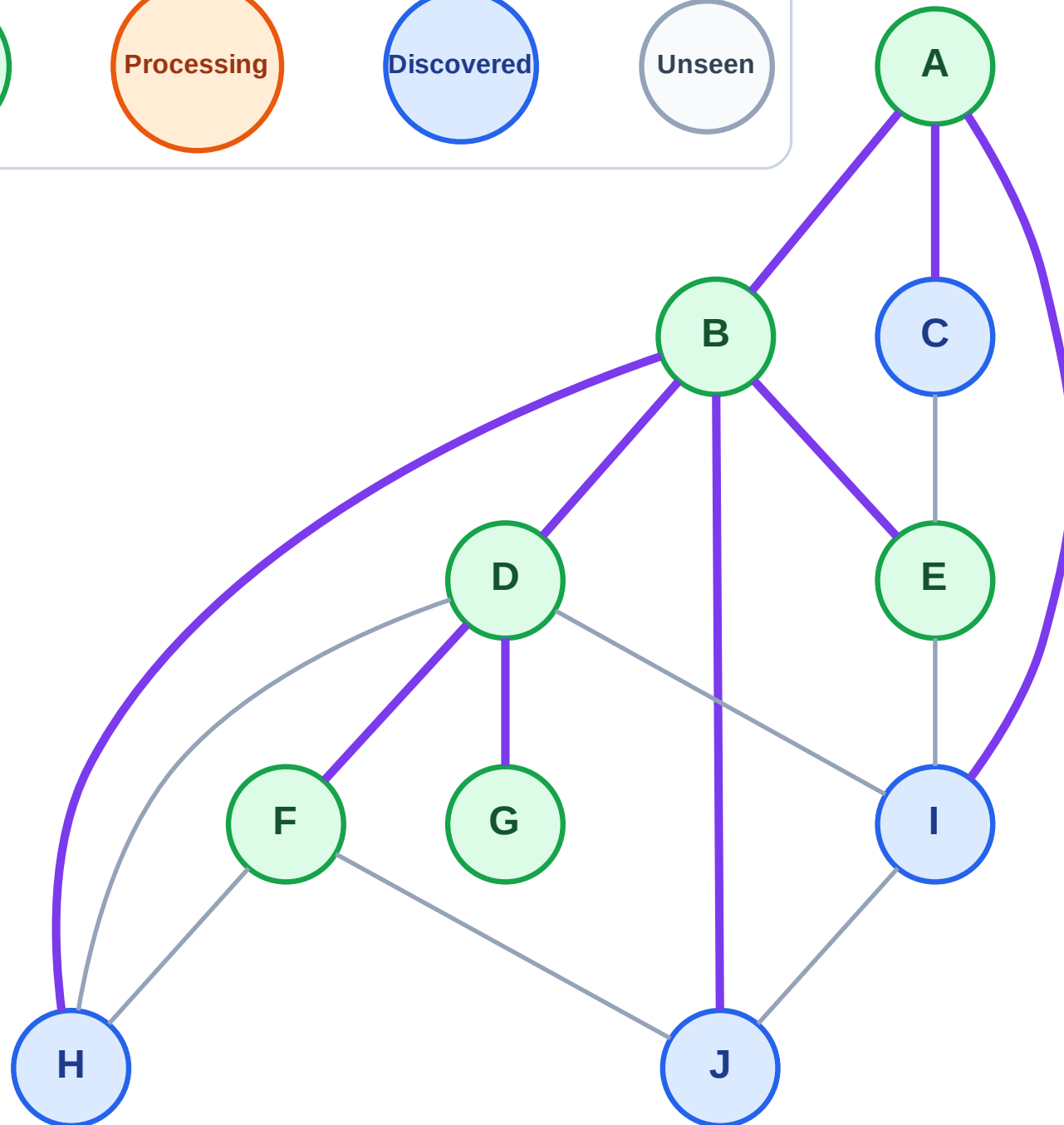
Step 13: No new neighbors from E

Legend



Stack (top at right): I | C | J | H

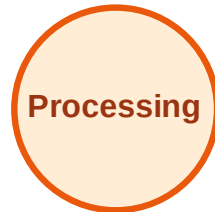
Visited: A, B, D, E, F, G



DFS Traversal

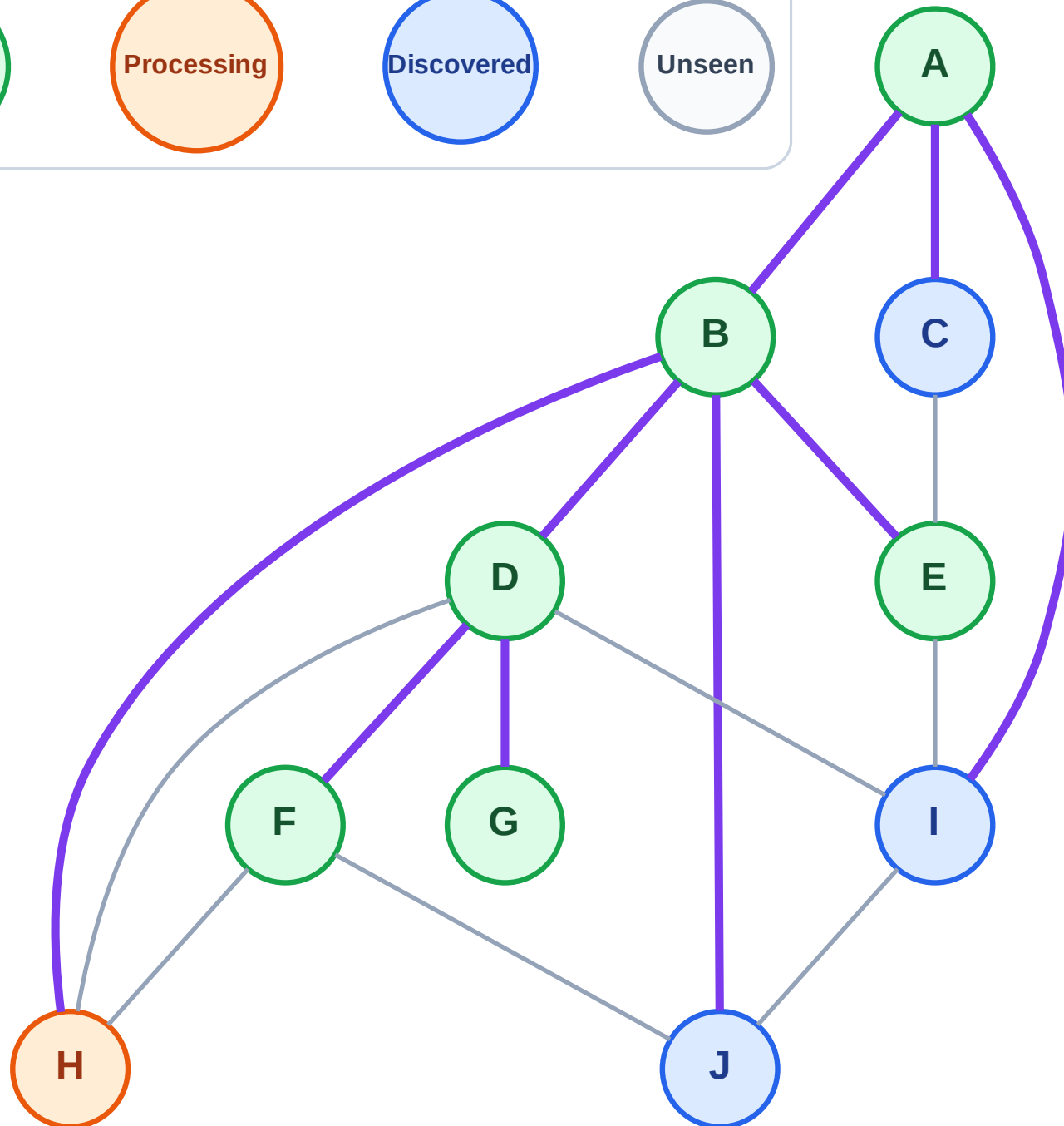
Step 14: Process node H

Legend



Stack (top at right): I | C | J

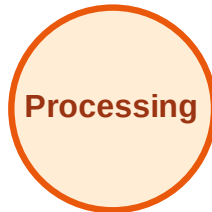
Visited: A, B, D, E, F, G



DFS Traversal

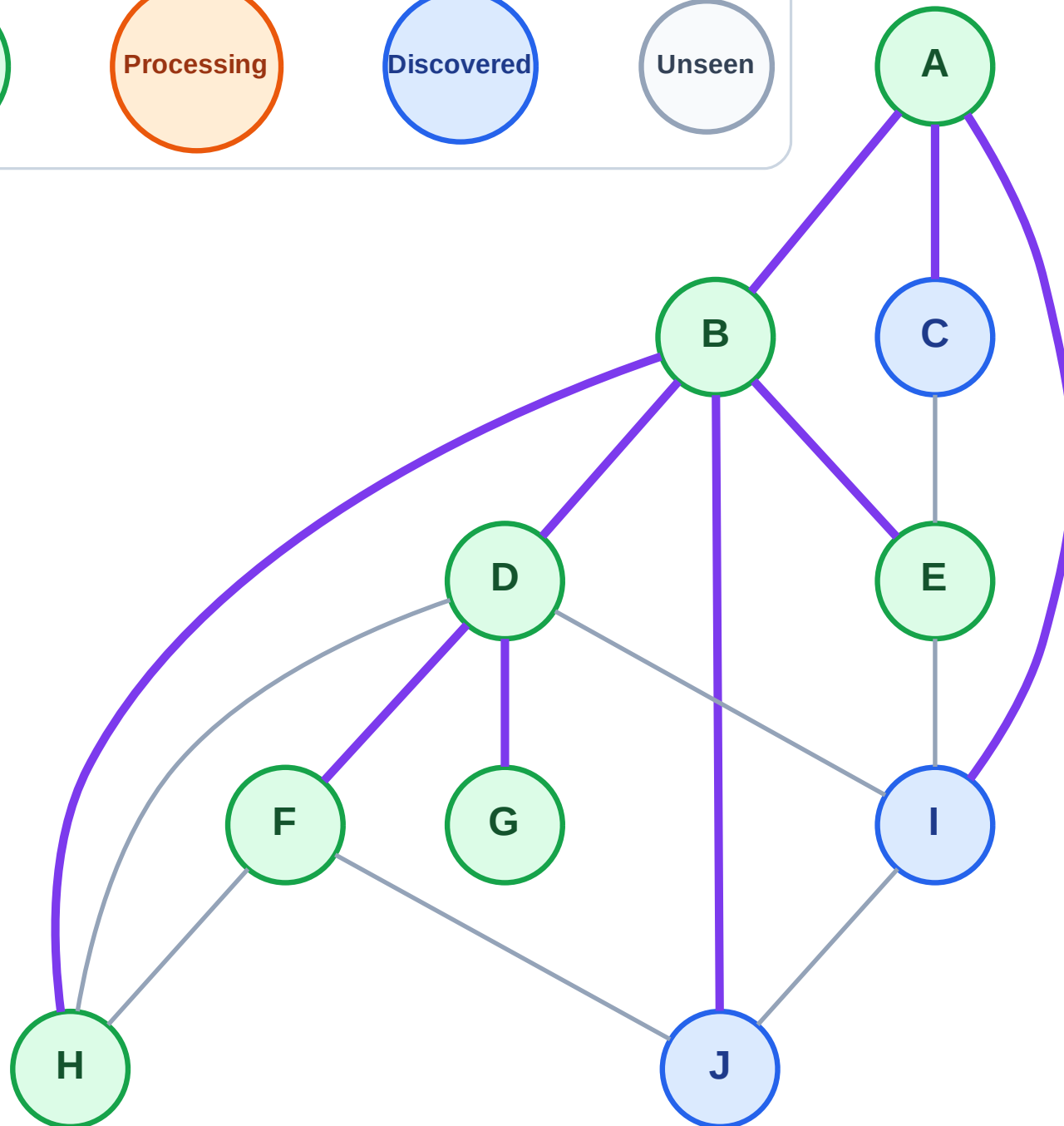
Step 15: No new neighbors from H

Legend



Stack (top at right): I | C | J

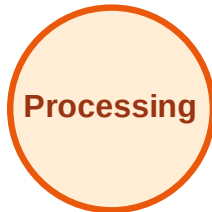
Visited: A, B, D, E, F, G, H



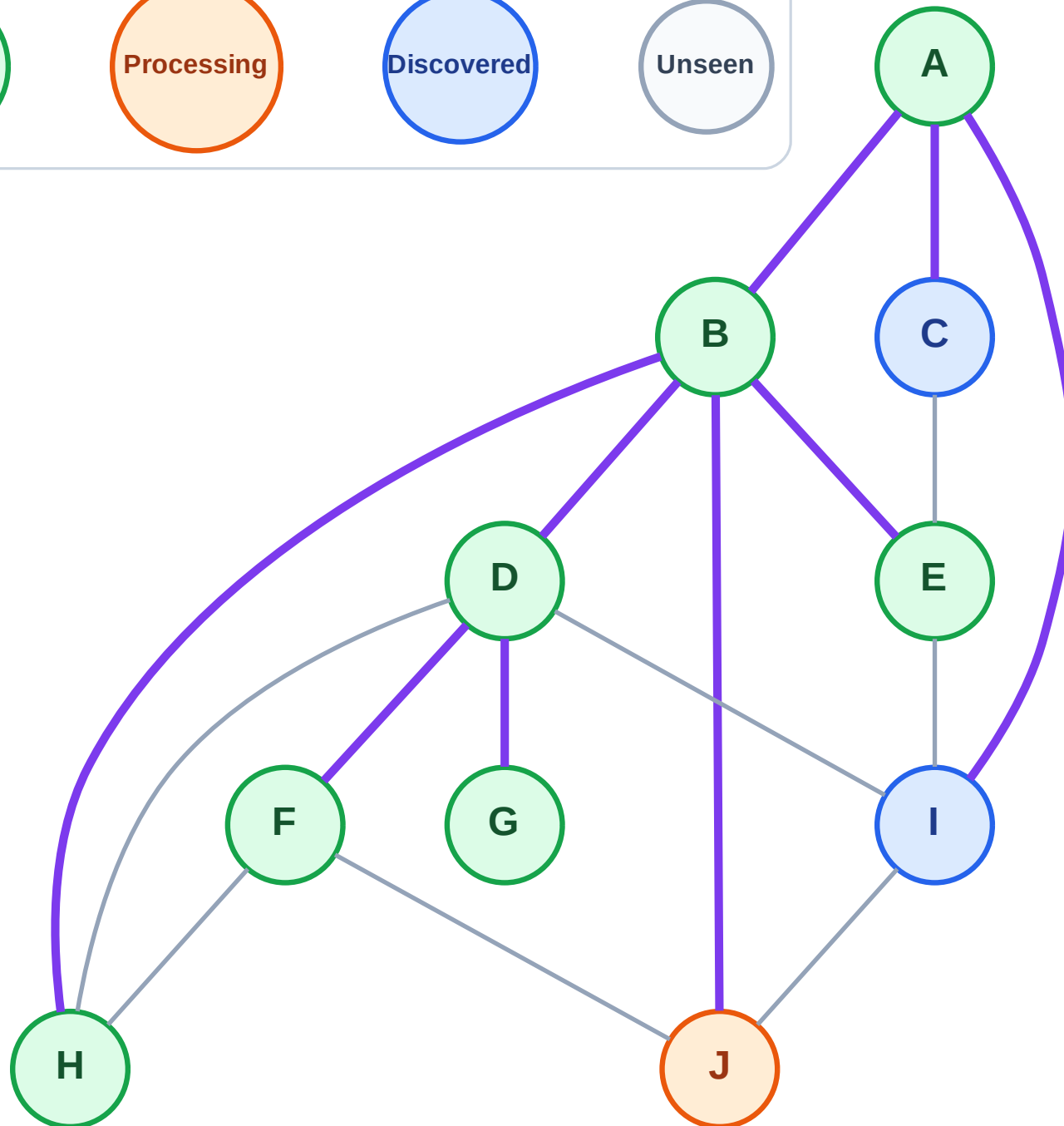
DFS Traversal

Step 16: Process node J

Legend



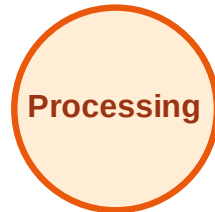
Stack (top at right): I | C
Visited: A, B, D, E, F, G, H



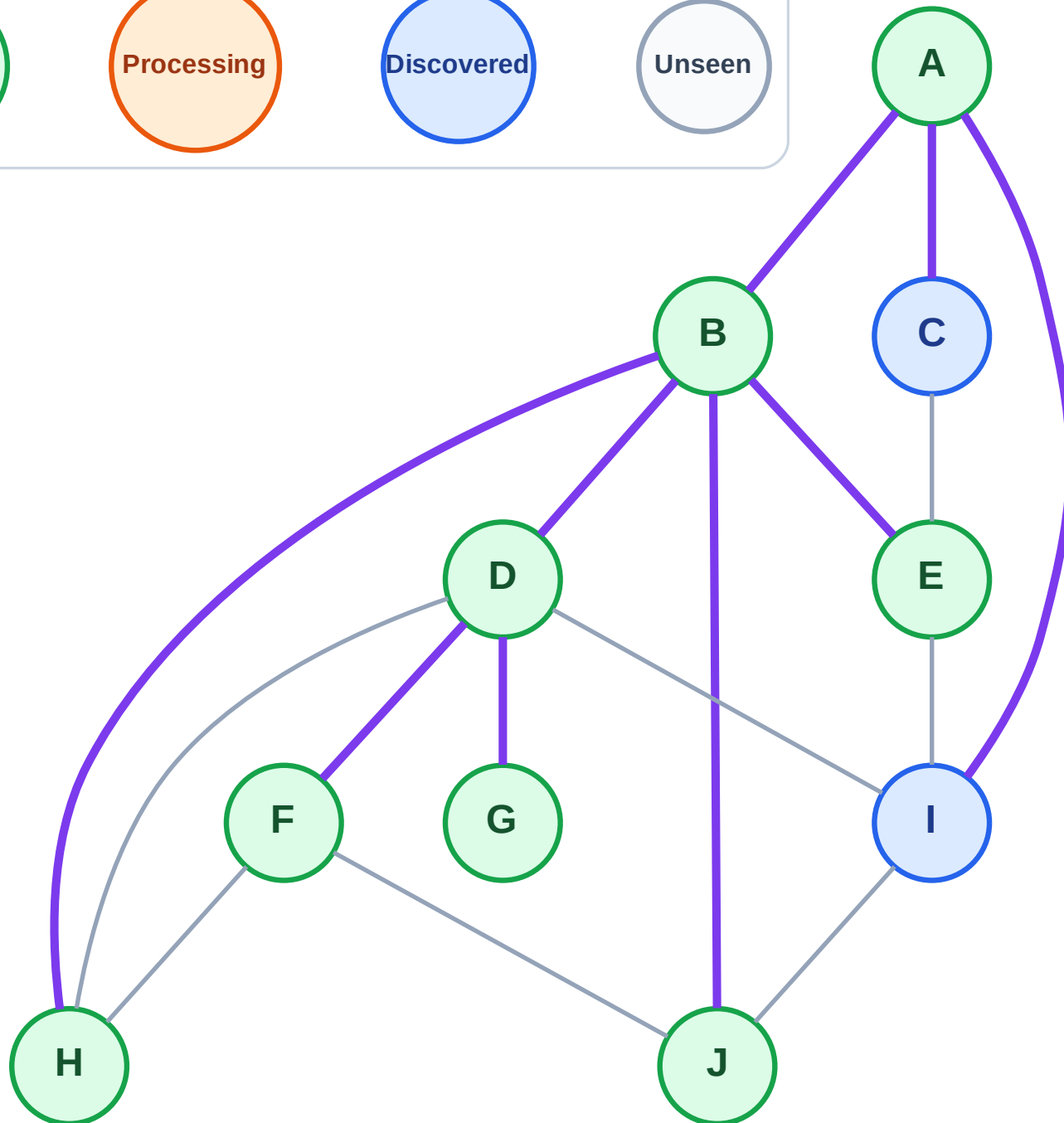
DFS Traversal

Step 17: No new neighbors from J

Legend



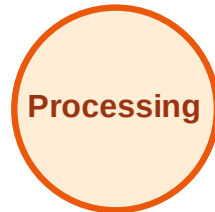
Stack (top at right): I | C
Visited: A, B, D, E, F, G, H, J



DFS Traversal

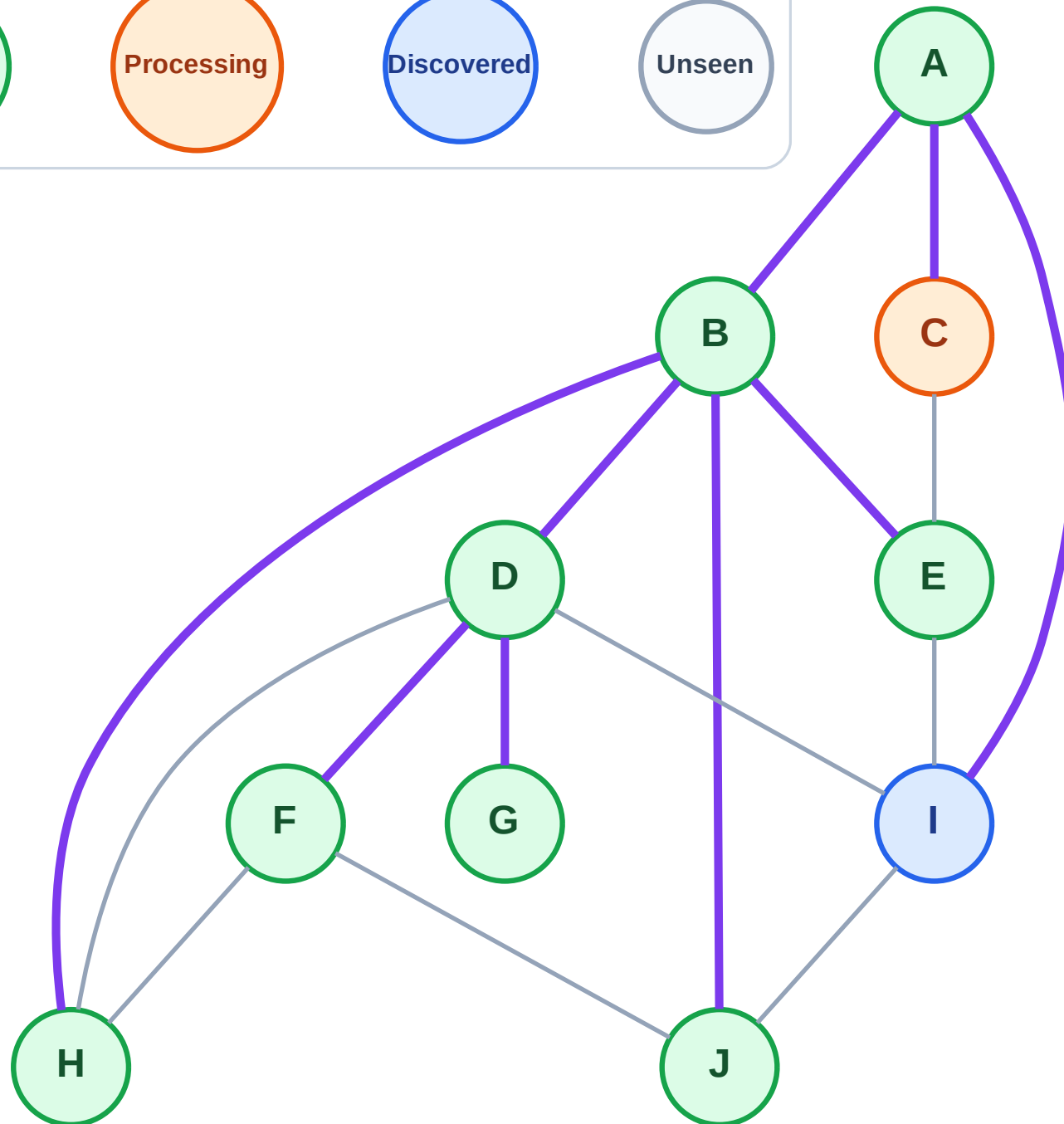
Step 18: Process node C

Legend



Stack (top at right): I

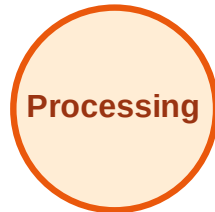
Visited: A, B, D, E, F, G, H, J



DFS Traversal

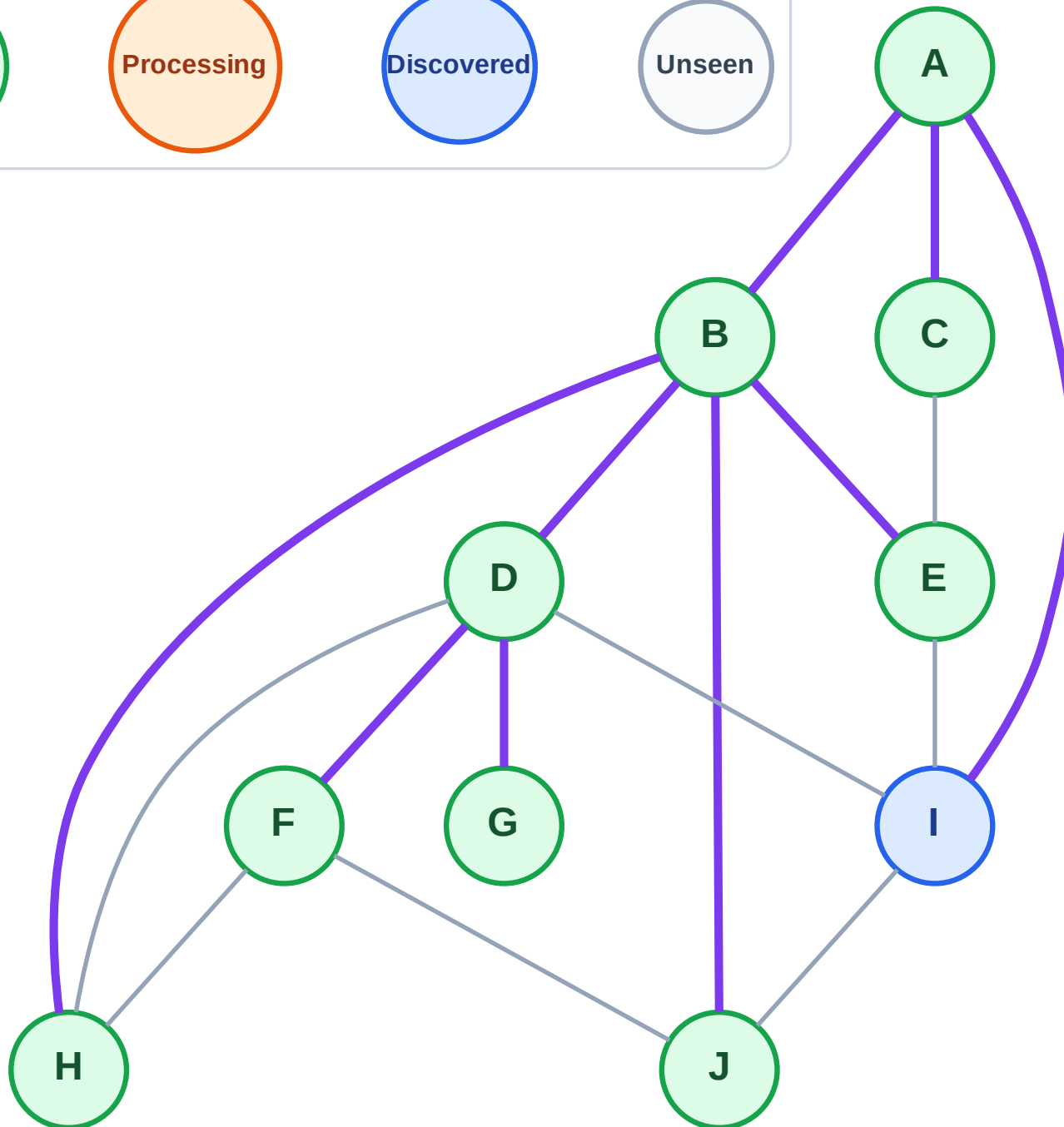
Step 19: No new neighbors from C

Legend



Stack (top at right): I

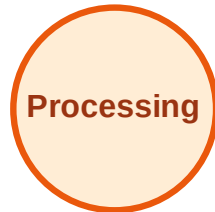
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

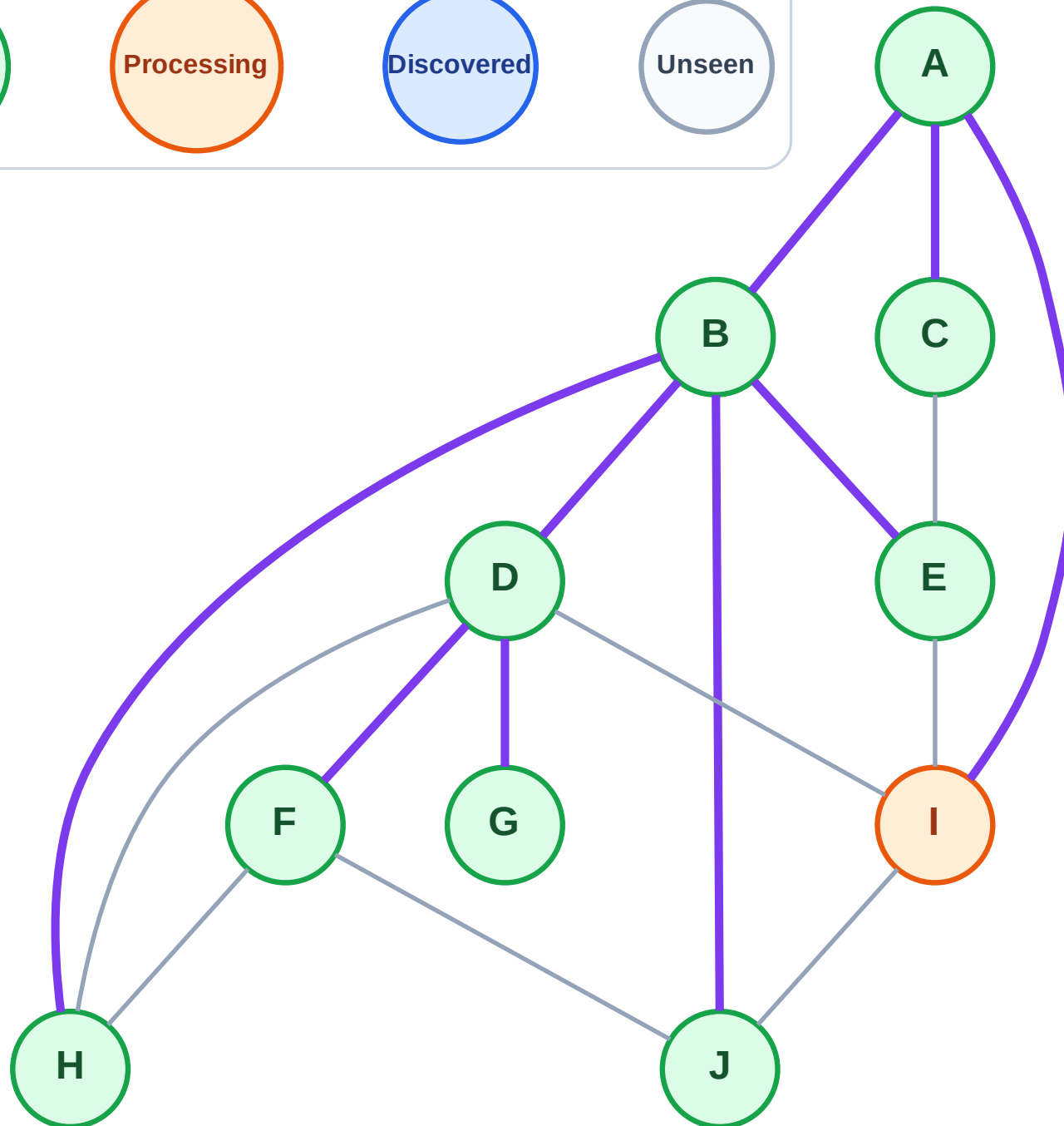
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

